MM-Net: A Multimodal Multi-Task Network for Sleep Staging and Sleep Apnea Assessment After Ischemic Stroke

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ABSTRACT

Automated polysomnography analysis after ischemic stroke presents distinct challenges because cerebrovascular injury can alter sleep-related neural activity, while respiratory disturbances introduce additional physiological complexity. Existing stroke-specific studies have primarily addressed sleep staging, leaving the joint characterization of sleep architecture and respiratory events insufficiently explored. We propose MM-Net, a multimodal multi-task framework that integrates neural and cardiorespiratory representations for concurrent sleep-stage classification and respiratory-event detection. The neural stream combines physiological descriptors with a sleep-adapted pretrained EEG encoder, while cross-fitting and embedding alignment address representation mismatch between downstream training and inference. A complementary cardiorespiratory stream learns directly from respiratory and cardiac waveforms, with shared temporal modeling and task-specific prediction pathways. On iSLEEPS, comprising 99 patients with subacute ischemic stroke, patient-independent ten-fold cross-validation yields 76.3% sleep-staging accuracy (Cohen's $\kappa = 0.656$) and respiratory-event AUC of 0.825, outperforming the evaluated baselines under matched protocols. On the independent ISRUC-I cohort, MM-Net achieves 80.3% staging accuracy and 0.889 respiratory AUC. External evaluations further demonstrate respiratory-event discrimination across different recording cohorts. The framework additionally supports patient-level apnea–hypopnea index estimation from respiratory scores. Controlled analyses identify distinct contributions of neural and cardiorespiratory signals, supporting joint physiological modeling for automated stroke polysomnography assessment.

1. Introduction

Sleep disturbances are common after ischemic stroke and can persist through the subacute phase of recovery. Sleep-disordered breathing is particularly prevalent in this population and has been associated with poorer functional outcomes [1–4]. Polysomnography (PSG) provides a detailed view of both processes within the same overnight recording. Electroencephalographic (EEG), ocular, and muscular activity supports characterization of sleep state, whereas airflow, respiratory effort, electrocardiography (ECG), pulse, and peripheral oxygen saturation reflect the physiological changes associated with respiratory disturbance. Automated analysis of these signals therefore has the potential to characterize two clinically relevant aspects of post-stroke sleep from a common examination.

Most automated PSG analysis has addressed these tasks separately. Sleep-stage classification has progressed from convolutional and recurrent networks to attention-based models and large pretrained EEG representations [5–12]. Respiratory-event detection has developed in parallel around airflow, respiratory effort, oximetry, and cardiac signals. More recently, joint models have shown that sleep stages and respiratory events can be estimated within a common framework. Huttunen *et al.* [13], for example, evaluated simultaneous sleep-stage and respiratory-event scoring across 877 clinical recordings, and related multi-task systems have since been studied in other sleep populations [14]. These studies establish the feasibility of joint analysis, but their evidence comes primarily from suspected-apnea, elderly, or general sleep cohorts.

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The corresponding problem in stroke is less well characterized. Cerebrovascular injury can alter sleep-related neural activity, including spindles, slow-wave activity, and hemispheric EEG patterns [15–20]. Models developed on conventional sleep cohorts may therefore encounter a substantially different neural signal distribution when applied after stroke. Existing work on the iSLEEPS cohort has begun to examine automatic sleep staging in this setting [21, 22], but joint analysis of sleep state and respiratory disturbance has not been characterized in the same population.

A second issue concerns what multimodal models learn from the available physiology. Predictive performance from a combined signal set does not identify which modalities support each output. This distinction is particularly relevant when sleep staging and respiratory-event prediction share a representation: neural and cardiorespiratory signals may contribute differently to the two tasks, and a respiratory model may also derive much of its discrimination from airflow, a signal closely related to clinical event annotation. Establishing the role of each physiological source therefore requires controlled intervention on the signal streams and channels in addition to conventional benchmark comparisons.

We investigate these questions using the iSLEEPS dataset [21], comprising overnight PSG from patients with subacute ischemic stroke. We develop MM-Net, a two-stream multi-task model in which neural and cardiorespiratory signals are represented separately before being integrated by a shared temporal decoder. The neural stream combines physiologically defined descriptors with a pretrained EEG representation, LaBraM [9], adapted to sleep from EEG and ocular channels. Because an adapted encoder has seen its own training patients, its embeddings are integrated into the downstream model through cross-fitting and alignment to the deployed encoder. The cardiorespiratory stream learns directly from eight respiratory and cardiac waveforms. The shared decoder produces a five-class sleep-stage output and a respiratory-event score for each 30-s epoch, with an additional cardiorespiratory pathway to the respiratory head for task-specific information, and the respiratory scores over decoded sleep yield a patient-level AHI estimate. The evaluation is patient-independent and combines comparative benchmarking with matched single-task controls, representation analyses, modality and channel interventions, nested model selection, repeated training seeds, training and evaluation on an independent sleep-disorder cohort, and transfer without retraining to sleep-disorder, healthy and sleep-apnea cohorts.

The principal contributions of this study are threefold:

1. Representation-consistent integration of sleep-adapted EEG features. We address the mismatch arising when an EEG encoder adapted on training patients supplies representations to a downstream model evaluated on unseen patients. A cross-fitting and embedding-alignment procedure integrates out-of-sample neural representations into a common deployment space, enabling the downstream framework to benefit from sleep-specific adaptation without relying on in-sample encoder outputs during training.
2. Task-specialized multimodal modeling of post-stroke sleep physiology. We develop a unified framework for sleep-stage classification and respiratory-event detection that combines neural and cardiorespiratory representations through shared temporal modeling while retaining task-specific respiratory information. This design enables concurrent characterization of sleep architecture and respiratory disturbances in subacute ischemic stroke, with patient-level AHI estimation as an additional output.
3. Evidence for physiological specialization and cross-cohort applicability. Through controlled signal interventions and patient-independent evaluations, we examine the dependence of each prediction task on its physiological inputs and establish the performance of the integrated framework against matched baselines. Independent-cohort experiments further assess the transfer of the learned representations beyond the stroke population.

The remainder of the paper describes the datasets and the model, followed by the experimental evaluation, the physiological analyses, and the discussion.

2. Related Work

2.1. Automatic Sleep Staging

Automatic sleep staging has progressed from feature-based classifiers to deep models that learn temporal and spectral representations directly from PSG signals. DeepSleepNet combines convolutional feature extraction with recurrent sequence modeling [5], and SeqSleepNet formulates sleep staging as sequence-to-sequence classification across consecutive epochs [6]. AttnSleep uses multi-resolution convolution and attention to capture complementary temporal characteristics [7], and TinySleepNet reduces model complexity while retaining recurrent modeling of stage transitions [8]. U-Time approaches staging as dense time-series segmentation [23], whereas XSleepNet combines raw-signal and spectral representations within a common sequence model [24].

More recent work has expanded the range of temporal models and learned representations. SleepTransformer applies attention-based sequence modeling [25], L-SeqSleepNet extends modeling across longer sleep-stage sequences [26], and ZleepAnlystNet and FlexSleepTransformer investigate alternative training strategies and flexible channel configurations [27, 28]. State-space architectures based on Mamba have also been explored for sleep staging [29, 30]. In parallel, large pretrained EEG models such as LaBraM, CBraMod, EEGPT, and BIOT provide representations learned from substantially larger collections of neural recordings [9–12]. Such encoders are typically used frozen or fine-tuned end to end for a single task. Using a fine-tuned encoder as the input of a second model trained on the same patients raises the requirement addressed by stacked generalization and cross-fitting, namely that the second model be trained on out-of-sample first-stage outputs [31, 32].

Most of these developments have been established on healthy or general sleep datasets. Performance can change substantially when models are transferred across cohorts because acquisition protocols, channel configurations, patient characteristics, and sleep physiology differ between datasets [33]. This issue is especially relevant after stroke, where the neural patterns used in conventional sleep staging may themselves be altered by neurological injury.

2.2. Multimodal and Joint PSG Analysis

Clinical PSG contains several complementary physiological modalities, and a parallel line of work has examined how these signals can be incorporated into automatic sleep analysis. U-Sleep, for example, was developed to operate across different EEG and EOG derivations and multiple sleep cohorts [34]. Other multimodal approaches combine neural, ocular, muscular, cardiac, or respiratory signals through early fusion, modality-specific encoders, or higher-level feature fusion. Recent work has also extended representation learning across PSG modalities, including SleepFM [35]. Cross-scale representations, self-supervised learning, and standardized sleep benchmarks have further broadened the range of models evaluated across datasets [36–38].

Joint modeling of sleep stage and sleep-related disorders has received less attention than sleep staging alone. Huttunen *et al.* [13] trained a single network for simultaneous sleep-stage and respiratory-event scoring across 877 clinical recordings and examined input configurations ranging from pulse oximetry to a broader PSG montage. Multitask approaches combining sleep staging with apnea-related prediction have subsequently been investigated in other clinical populations [14]. These studies demonstrate that the two outputs can be learned within a common model, but they have primarily been evaluated in suspected-apnea, elderly, or general sleep populations.

An additional question in multimodal modeling is whether improved predictive performance reflects complementary information across modalities or strong dependence on a particular signal. This distinction is important when different outputs are associated with different physiological systems. EEG, EOG, and EMG contain information directly related to sleep state, whereas airflow, respiratory effort, oxygen saturation, and cardiac activity characterize different aspects of respiratory disturbance. Aggregate multimodal performance alone does not establish how these signal groups contribute to individual outputs.

2.3. Respiratory-Event Detection

Automated analysis of sleep-disordered breathing has largely developed as a separate task. Apnea and hypopnea detectors commonly use airflow, thoracic and abdominal respiratory effort, oxygen saturation, and ECG-derived cardiac measures. These signals describe different stages of the physiological response: airflow reflects reduction or cessation of ventilation, respiratory effort provides information about continued or absent breathing effort, and oximetry captures the subsequent change in blood oxygenation. Oxygen desaturation also forms part of the clinical criteria used in hypopnea scoring [39]. End-to-end deep models have also been developed for respiratory-event scoring, including detection from a single respiratory-effort belt [40]. Biswal *et al.* [41] trained a recurrent–convolutional network to jointly score sleep stages, apnea events, and limb movements from clinical PSG.

Respiratory-event models are typically optimized independently of sleep staging and often operate on temporal resolutions different from the 30-s epoch used for stage classification. Joint models reduce this separation by learning both outputs from the same recording, but the interpretation of the respiratory output requires particular care when the model has access to signals involved in event annotation. Evaluating the contribution of individual respiratory signals therefore provides information that overall discrimination metrics alone cannot supply.

2.4. Sleep and Sleep-Disordered Breathing After Stroke

Sleep physiology is frequently altered after stroke, and sleep disturbances have been associated with neurological and functional recovery [1–3]. Changes in sleep-related neural activity have also been reported after cerebrovascular

Table 1
Datasets used in the study and their roles in the evaluation.

Dataset	Population	Data used	Role
iSLEEPS [21]	Subacute ischemic stroke	99 patients	Development and in-domain evaluation
Sleep-EDF [42, 43]	Healthy sleep	40 recordings	Reimplementation checks, healthy-sleep reference, and external staging
ISRUC-Sleep I [44]	Sleep disorders	100 patients	Cross-validation and transfer (staging, respiratory, AHI)
ISRUC-Sleep II [44]	Sleep disorders	8 subjects, two nights each	Cross-validation and transfer
ISRUC-Sleep III [44]	Healthy	10 subjects	Cross-validation and transfer
UCD [45]	Suspected sleep apnea	25 patients	Transfer (staging, respiratory, AHI)

injury. Previous studies describe alterations in sleep spindles, slow-wave activity, and related electrophysiological patterns [15–19]; quantitative EEG measures such as the brain symmetry index have also been examined in relation to hemispheric injury [20]. These changes make stroke sleep a distinct setting for models whose representations were developed predominantly on non-stroke populations.

Sleep-disordered breathing is also common during the subacute phase, so sleep state and respiratory disturbance coexist within the same clinical recording. Despite this overlap, automated stroke-specific sleep analysis remains limited. The iSLEEPS dataset introduced a dedicated PSG resource for subacute ischemic stroke and reported baseline sleep-staging performance [21]; subsequent work has further examined the behavior of deep staging models on this cohort [22]. Existing stroke-specific studies have therefore concentrated primarily on sleep-stage classification. Joint epoch-level modeling of sleep stage and respiratory disturbance, together with controlled analysis of the neural and cardiorespiratory information supporting each output, remains insufficiently characterized in this population.

3. Datasets and Preprocessing

Four publicly available polysomnography datasets are used in this study (Table 1): iSLEEPS [21], Sleep-EDF [42, 43], ISRUC-Sleep [44] with its three subgroups, and the St Vincent's University Hospital / University College Dublin sleep apnea database (UCD) [45]. iSLEEPS is the primary cohort used for model development and patient-independent evaluation. Sleep-EDF is used as a healthy-sleep reference for implementation checks, transfer experiments, and subject-independent evaluation of the staging model outside the stroke cohort. ISRUC-Sleep provides independent sleep-disorder and healthy cohorts with both sleep-stage and respiratory-event annotations; it is used both to train and evaluate MM-Net under subject-independent cross-validation and to test the iSLEEPS-trained model without retraining. UCD provides a further sleep-apnea cohort recorded on a different acquisition system and is used for transfer only. No external cohort is used to select the reported iSLEEPS model configuration.

3.1. iSLEEPS: Primary Stroke Cohort

The iSLEEPS dataset contains overnight PSG from 100 patients with subacute ischemic stroke [21]. The released cohort has a mean age of 50.5 years and includes 77 male and 23 female participants. Sleep is scored into the five AASM stages: Wake (W), N1, N2, N3, and rapid-eye-movement (REM) sleep. Respiratory events are also annotated throughout the recordings.

An audit performed before cross-validation identified the records of SN15 and SN28 as byte-identical. Retaining both identifiers could place the same recording in different patient-level folds. One member of the duplicate pair was therefore removed before constructing the evaluation splits, leaving $N = 99$ patients and 92,560 30-s epochs for the analyses reported here.

The resulting sleep-stage distribution is shown in Table 2. N2 accounts for the largest proportion of epochs, and N1 and N3 are the least frequent stages. Respiratory-event onsets are marked in 16.0% of all epochs. At the patient level,

Table 2
Sleep-stage distribution in the analyzed iSLEEPS cohort ($N = 99$; 92,560 epochs).

Stage	W	N1	N2	N3	REM
Share (%)	26.7	10.1	42.3	8.9	12.1

the fraction of event-positive epochs ranges from 0 to 60.5%; 80 of the 99 patients have respiratory events in more than 5% of their epochs, and the median burden is 13% (Figure 1).

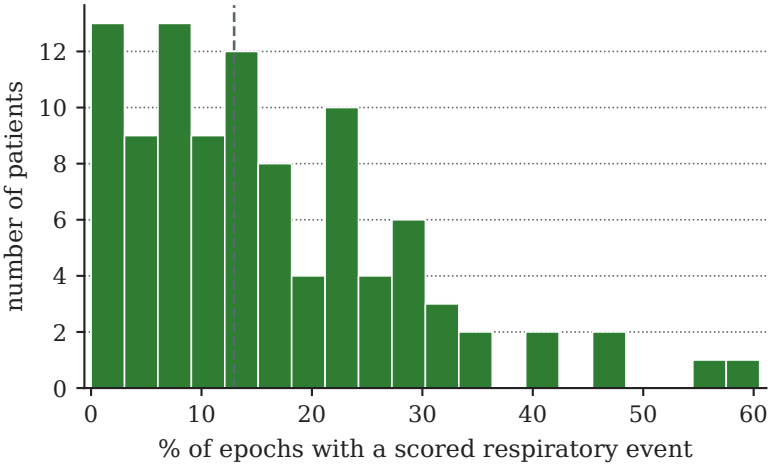


Figure 1: Patient-level respiratory-event burden in iSLEEPS, measured as the fraction of 30-s epochs containing the onset of a scored respiratory event. The dashed line denotes the cohort median of 13%.

3.2. Sleep-EDF: Healthy-Sleep Reference Cohort

Sleep-EDF is used to provide a healthy-sleep reference [42, 43]. The experiments use the Sleep Cassette subset corresponding to subjects 0–20, comprising 40 overnight recordings. Because individual subjects can contribute more than one night, splits are constructed at the subject level.

Sleep-EDF serves three purposes in this study. First, it provides an independent check of the reimplemented sleep-staging architectures against the healthy-sleep setting in which those models were originally evaluated. Second, models trained on Sleep-EDF are applied to iSLEEPS without adaptation to examine transfer from healthy sleep to the stroke cohort. Third, the proposed staging model is evaluated on Sleep-EDF as an independent healthy-sleep cohort. The database provides frontal EEG derivations including Fpz-Cz and Pz-Oz. It does not provide the full cardiorespiratory signal set required by the proposed respiratory branch and contains no respiratory-event annotations used by this study; consequently, respiratory performance is not evaluated on Sleep-EDF.

3.3. ISRUC-Sleep: Independent Sleep-Disorder Cohort

ISRUC-Sleep provides the independent evaluation in populations with and without sleep disorders [44]. Subgroup I contains 100 patients with sleep disorders recorded on one night each, subgroup II eight patients recorded on two nights each, and subgroup III ten healthy adults. The recordings include central and occipital EEG derivations corresponding to C4, C3, O2, and O1 referenced to the contralateral mastoid, two EOG derivations and chin EMG, together with oronasal airflow, nasal pressure, thoracic and abdominal effort, SpO_2, and ECG, and annotations of obstructive, central, and mixed apneas and hypopneas; the annotations of the first scorer are used. Summed effort is formed as the sum of the thoracic and abdominal traces, and pulse as the beat-to-beat heart rate derived from ECG R-peaks, so that every iSLEEPS input has an ISRUC counterpart. In 30 of the 126 ISRUC recordings the EDF header declares the EEG and EOG range as $\pm 25\ \mu$V, which places their N3 peak-to-peak amplitude at 9–23 μV against 98–283 μV in the remaining recordings and below the 75 μV that defines N3; these channels are restored by a factor of ten, a correction decided from the header alone. One subgroup-I recording stores monopolar EEG and EOG, from which the mastoid-referenced derivations are formed.

ISRUC-Sleep therefore supports evaluation of every output of the proposed framework: sleep-stage prediction, respiratory-event prediction and, in subgroup I, AHI estimation across the full severity range.

3.4. UCD: Independent Sleep-Apnea Cohort

The UCD database [43, 45] contains overnight PSG from 25 adults referred for suspected sleep-disordered breathing, with annotated AHI between 1.2 and 81.2 events/h. It records C3 and C4 EEG only, which also supply the occipital inputs, two EOG channels, chin EMG, ECG, oronasal airflow, thoracic, abdominal and summed effort, SpO_2 and pulse, and its scored apneas and hypopneas carry onset times. The EDF headers of this database carry no physical unit, so EEG, EOG and EMG are scaled per recording to the median whole-night amplitude of iSLEEPS, using no labels. UCD is used only for transfer.

3.5. Signal Selection and Harmonization

The iSLEEPS recordings contain 22 physiological and auxiliary traces. Fifteen signals are retained for modeling and divided into neural and cardiorespiratory groups. The neural group contains four EEG derivations (C4:M1, C3:M2, O2:M1, and O1:M2), two EOG derivations (E1:M2 and E2:M2), and the submental EMG. The cardiorespiratory group contains ECG, oronasal thermal airflow, nasal pressure, thoracic effort, abdominal effort, summed respiratory effort, SpO_2, and pulse.

Signals associated with periodic limb movement, snoring, plethysmography, body position, ambient light, and device operation are not included in the model. The snore channel is additionally excluded from respiratory prediction because it is closely related to the airway information used during respiratory-event assessment. Twelve iSLEEPS recordings lack at least one selected cardiorespiratory channel. These patients are retained, and unavailable inputs are zero-filled. The same convention is used when a selected signal is unavailable in an external dataset.

The datasets differ in montage and in the availability of cardiorespiratory channels. Dataset-specific signal availability is therefore preserved, and external recordings are represented according to their available channels without assuming the complete iSLEEPS montage. Sleep-EDF is used only for analyses supported by its available sleep-staging signals, whereas ISRUC-Sleep provides sufficient respiratory information to evaluate the respiratory output with unavailable inputs explicitly represented as missing channels.

3.6. Preprocessing and Epoch Labels

All sleep-stage predictions are defined on 30-s epochs. For iSLEEPS and ISRUC-Sleep, the respiratory annotations are mapped to the same epoch grid. An epoch is assigned a positive respiratory label if a scored respiratory event begins within it and a negative label otherwise. The primary respiratory task is therefore prediction of respiratory-event onset within a 30-s epoch, and the patient-level AHI is derived from these predictions (Section 4.8).

iSLEEPS retains only the epochs scored as a sleep stage, so a recording whose hypnogram contains unscored rows has fewer retained epochs than 30-s slots on the recording clock. Every cardiorespiratory signal and every respiratory-event label is therefore cut at the true recording time of each retained epoch. Cutting them on the clock grid instead would shift them against the EEG after the first unscored row; in iSLEEPS this concerns 63 of the 100 released recordings, by up to 137 epochs. The annotated AHI is the number of apneas and hypopneas starting in scored sleep per hour of scored sleep.

For the iSLEEPS neural representation, the selected EEG, EOG, and EMG signals are resampled to 100 Hz for computation of the engineered descriptors described in Section 4.2. The LaBraM encoder receives the four EEG derivations and the two EOG derivations. Because LaBraM was pretrained at 200 Hz, these signals are independently resampled to 200 Hz before entering the encoder. The EMG channel is not supplied to LaBraM.

The eight cardiorespiratory inputs are resampled to a common 25 Hz grid using polyphase resampling before entering the learned cardiorespiratory encoder. SpO_2 is recorded natively at 4 Hz and is placed on the same 25 Hz grid for temporal alignment with the remaining channels; this operation does not increase the underlying temporal resolution of the oximeter signal. Raw cardiorespiratory signals remain in their recorded physical units after resampling.

Engineered feature representations are standardized within each recording before model input. For iSLEEPS, where each analyzed patient contributes one overnight recording, this corresponds to the patient-wise standardization used throughout the primary analysis. The same per-recording procedure is applied when the trained model is evaluated on the external datasets. No labels from an external dataset are used to refit model parameters, select thresholds, or adapt the reported iSLEEPS model.

4. Proposed Method

4.1. Problem Formulation

A night is a sequence of T epochs. For epoch t we form a neural feature vector $\mathbf{f}_t^{\text{eeg}} \in \mathbb{R}^{388}$, formed by concatenating the 188 engineered descriptors below with a 200-dimensional embedding from LaBraM, a pretrained EEG encoder [9] adapted to sleep (Section 4.3), and a raw cardiorespiratory tensor $\mathbf{x}_t^{\text{car}} \in \mathbb{R}^{8\times750}$, the eight channels at 25 Hz. The 14 engineered cardiorespiratory descriptors defined below are used by the baseline configuration against which the learned encoder is measured.

The pretrained encoder is supplied with input in the format used during its pretraining. LaBraM was pretrained at 200 Hz and fixes its input length at 3000 samples, so the EEG is resampled from our 100 Hz working rate to 200 Hz before it reaches the encoder. Supplying the 100 Hz signal directly would introduce an undetected error: 3000 samples would then span 30 s of recording while the encoder interprets them as 15 s, compressing time by a factor of two, so that every rhythm would appear at twice its frequency and a 12 Hz spindle at 24 Hz. At 200 Hz a 30 s epoch no longer fits the positional embedding in one piece, so each epoch is split into two 15 s sub-windows of 3000 samples, encoded separately, and mean-pooled; the two embeddings were verified to differ before caching, so the pooling combines distinct information from the two sub-windows. The encoder also requires canonical 10–20 names and rejects mastoid references, so C4:M1, C3:M2, O2:M1 and O1:M2 are presented as C4, C3, O2 and O1. The two EOG derivations are presented as F7 and F8, the frontal positions nearest the eyes, so that the encoder receives the ocular activity that separates REM from Wake; the chin channel is not supplied. Amplitudes are scaled to μV/100. The encoder therefore sees four EEG and two EOG channels. We learn a subject-independent map

$$f_\theta : \left(\mathbf{f}_{1:T}^{\text{eeg}}, \mathbf{x}_{1:T}^{\text{car}}\right) \longmapsto \left(\hat{y}_{1:T}^{\text{stg}}, \hat{y}_{1:T}^{\text{apn}}\right), \tag{1}$$

where $\Delta^4 = \{p \in \mathbb{R}^5 : p_k \geq 0, \sum_k p_k = 1\}$ is the four-simplex, $\hat{y}_t^{\text{stg}} \in \Delta^4$ is a distribution over the five stages and $\hat{y}_t^{\text{apn}} \in [0, 1]$ scores whether epoch t contains a respiratory event (Figure 2). We write $\hat{y}_t^{\text{apn}}$ as a score and not as a probability for the reason given in Section 4.7: the positive weighting in its loss displaces the optimum away from the conditional event rate.

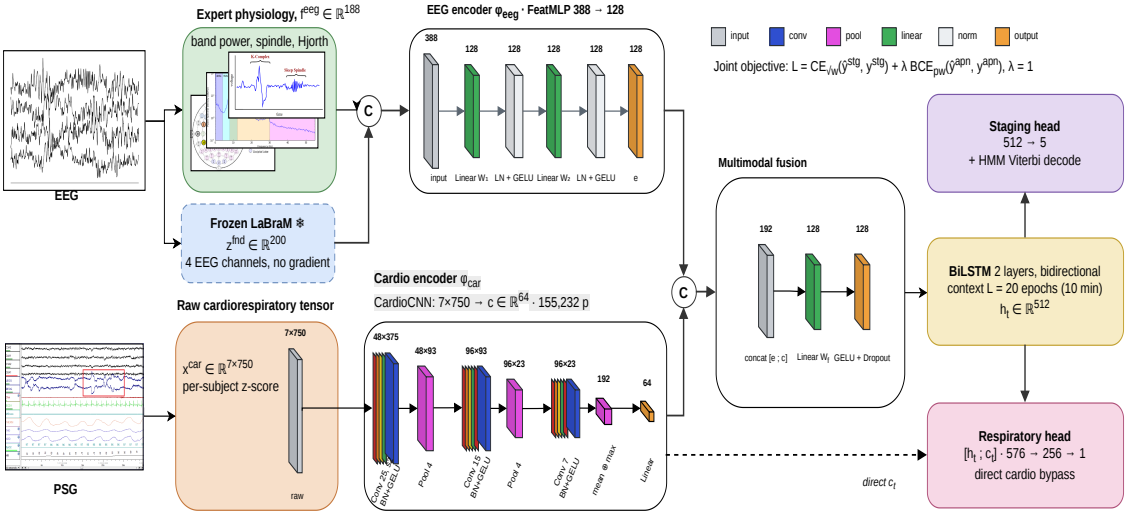


Figure 2: The two-stream, multi-task model. The neural stream combines 188 engineered descriptors with the embedding of a LaBraM encoder adapted to sleep from EEG and EOG and integrated by cross-fitting and alignment; the cardiorespiratory stream is a convolutional encoder over eight raw channels. The streams are fused and read by a bidirectional LSTM over a 20-epoch window, which feeds a staging head and a respiratory head; the respiratory head also reads the cardiorespiratory embedding directly, and the AHI is computed from the respiratory scores over decoded sleep.

4.2. Physiological Feature Representations

*Neural features (*188*)* Per channel we take absolute and relative band powers from the Welch spectrum $P(f)$ in delta (0.5–4), theta (4–8), alpha (8–13), sigma (11–16), and beta (16–30) Hz; the relative power in band b is

$$\mathrm{RP}_b = \frac{\int_{f\in b} P(f)\,df}{\int_{0.5}^{30} P(f)\,df}. \tag{2}$$

The bands are the conventional sleep bands and alpha and sigma overlap between 11 and 13 Hz, so the five relative powers are not a partition and do not sum to one. We retain the conventional definitions because both bands correspond to physiology assessed in visual scoring, and the model treats the five values as correlated descriptors whose sum is unconstrained. We add spectral entropy $H = -\sum_i p_i \log p_i$ with p_i the normalized spectrum, the 95% spectral-edge frequency, the three Hjorth descriptors [46], and time-domain statistics. Sleep spindles are detected on the sigma-band analytic envelope, the modulus of the analytic signal built from the Hilbert transform $\mathcal{H}$, $e(t) = |x_\sigma(t) + i\,\mathcal{H}\{x_\sigma\}(t)| = \sqrt{x_\sigma(t)^2 + \mathcal{H}\{x_\sigma\}(t)^2}$, with density counted above a per-epoch threshold. No duration, separation or morphology criterion is applied, so this measure counts crossings of a sigma-band envelope and serves as a proxy for spindle density; it is not a validated spindle detector. The slow-wave descriptors likewise summarise slow-wave amplitude without detecting discrete waves. Ocular-movement energy on the EOG and tonic EMG level complete the set.

*Cardiorespiratory features (*14*)* From the seven cardiorespiratory channels we compute SpO_2 mean, minimum, variability, and desaturation depth (median minus minimum); pulse mean and variability; ECG amplitude and line-length; airflow amplitude and line-length; thoracic, abdominal, and summed effort amplitudes; and signed thoraco-abdominal synchrony, whose loss is the paradoxical motion of chest against abdomen that marks an obstructed breath. The correlation is kept signed, so synchronous and paradoxical motion sit at opposite ends of the range; a value near +1 is synchronous motion and a negative value is paradox,

$$\mathrm{sync}_t = \mathrm{corr}(x_t^{\mathrm{tho}},\ x_t^{\mathrm{abd}}). \tag{3}$$

Both feature sets are standardized per patient, which removes person-to-person offsets and reduces the sensitivity of the model to inter-individual differences.

The learned cardiorespiratory encoder The evaluated model does not read the 14 descriptors. Its cardiorespiratory branch is a three-layer convolutional encoder over the raw 8×750 tensor, with wide kernels and aggressive downsampling because respiratory events are slow: at 25 Hz the first kernel of 25 samples spans one second, and a hypopnea unfolds over 10 to 30 s. Channel widths are 48, 96 and 96, each followed by batch normalisation and GELU, with max-pooling by four after the first two. Mean and max pooling over time are concatenated and projected by a linear layer, LayerNorm, GELU and dropout to the same 64 dimensions as the cardiorespiratory feature MLP it replaces, so the encoder is a drop-in replacement and every other part of the network is unchanged. The branch holds about 0.16 M trainable parameters (155,232 in the seven-channel configuration). Section 5.15 quantifies the effect of this substitution, which is among the largest representation changes we measure, and Section 5.2 shows that the effect is attributable to the change of representation: random encoders of comparable capacity over the same raw signal recover much of it. This encoder is used because it is small, trained end-to-end with the rest of the network, and adds no external dependency.

4.3. Sleep-Adapted EEG Encoder

Adaptation Within each training fold, LaBraM-Base (5.8 M parameters) is fine-tuned with a linear staging head on single epochs of the fold's training patients, using AdamW at 3×10^{-5} for the encoder and 10^{-3} for the head, a one-cycle schedule over three passes of the training epochs, batch size 64 and mixed precision; the state with the highest accuracy on the fold's validation patients is kept. Test patients never reach the encoder.

Cross-fitting An encoder fine-tuned on a patient embeds that patient in-sample. If the downstream model is trained on such embeddings and tested on out-of-sample ones, the two distributions differ, and the downstream model is fitted to representations of higher quality than those available at test time. Each fold's training patients are therefore split into $K = 3$ parts P_1, P_2, P_3, and an encoder ϕ_k is fine-tuned on the training patients outside P_k; every training patient in P_k is embedded by ϕ_k, an encoder that never saw it. The full-fold encoder ϕ, fine-tuned on all training patients, embeds the validation and test patients.

Alignment Cross-fitting makes every embedding out of sample but places the training patients in the spaces of K different encoders, which do not coincide with the space of ϕ used at test time. Each ϕ_k is mapped into the space of ϕ by a ridge regression [47] fitted on the patients of P_k, for whom both embeddings exist,

$$W_k = \arg\min_W \left\| (E^{\phi}_{P_k} - \mu^{\phi}_k) - (E^{\phi_k}_{P_k} - \mu^{\phi_k}_k) W \right\|^2_F + \lambda n_k \|W\|^2_F, \tag{4}$$

and the training embeddings become $\tilde{\mathbf{e}}_t = (\mathbf{e}^{\phi_k}_t - \mu^{\phi_k}_k) W_k + \mu^{\phi}_k$, with $\lambda = 0.01$ and n_k the number of epochs in P_k. The map uses training patients only. Every training embedding is thereby out of sample and expressed in the coordinates the downstream model receives at test time; Section 5.16 shows that both properties are required.

4.4. Two-Stream Encoders

The neural stream is a two-layer perceptron with LayerNorm; the cardiorespiratory stream is the convolutional encoder described above, which emits a vector of the same width as the feature perceptron it replaced,

$$e_t = \mathrm{GELU}\big(\mathrm{LN}(W_2\,\mathrm{GELU}(\mathrm{LN}(W_1 \mathbf{f}^{\mathrm{eeg}}_t)))\big) \in \mathbb{R}^{128}, \tag{5}$$

$$c_t = \mathrm{CardioCNN}\big(\mathbf{x}^{\mathrm{car}}_t\big) \in \mathbb{R}^{64}. \tag{6}$$

LayerNorm is used in place of BatchNorm because the recurrent windows are short and mix patients within a batch, and the cardiorespiratory stream is deliberately narrower because the branch compresses an 8×750 tensor into the same 64 dimensions as the feature encoder it replaced, so that the substitution is the only variable.

4.5. Fusion

Two fusion operators are used. Concatenation projects the joined embeddings back to the stream width,

$$z_t = \mathrm{GELU}\big(W_f\,[\,e_t\,;\,c_t\,]\big) \in \mathbb{R}^D, \qquad D = 128. \tag{7}$$

Cross-attention treats the two embeddings as tokens carrying learned modality-type embeddings, refines them by four-head scaled dot-product attention so that each modality reads the other, and projects the result to $z_t \in \mathbb{R}^D$. In the feature-only configuration the two were indistinguishable on staging (+0.005 accuracy, $p = 0.49$) and concatenation was slightly ahead on the respiratory head (0.014 AUC, $p = 0.06$), so the controlled analyses below use concatenation. With uniform class weights and the sleep-adapted encoder, cross-attention was carried forward on validation accuracy and is the fusion of the final MM-Net (Table 13).

4.6. Temporal Decoder and Two Heads

Sleep is sequential and respiratory events cluster, so the fused vectors of an $L = 20$ epoch window (10 minutes) are read by a two-layer bidirectional LSTM. Training draws windows at a stride of $L/2$, so an interior epoch is seen twice per pass and an epoch in the first or last ten of a recording once; no correction is applied for this imbalance, and the first and last window of each night are zero-padded to L under a mask that removes the padding from the loss. Inference uses disjoint windows: a recording is padded to a multiple of L, split into consecutive blocks, and the padding discarded, so every epoch is scored exactly once and no stitching rule is needed. Hidden states are not carried between windows in either case. Each direction runs the standard recurrence

$$\begin{aligned} i_t &= \sigma(W_i[z_t, h_{t-1}]), & f_t &= \sigma(W_f[z_t, h_{t-1}]), \\ o_t &= \sigma(W_o[z_t, h_{t-1}]), & g_t &= \tanh(W_g[z_t, h_{t-1}]), \\ s_t &= f_t \odot s_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(s_t), \end{aligned} \tag{8}$$

and the forward and backward states of the 256-unit layers are concatenated into $h_t \in \mathbb{R}^{512}$. The staging head is a linear map $\hat{y}^{\mathrm{stg}}_t = \mathrm{softmax}(W_s h_t)$. The respiratory head reads h_t together with a direct copy of the per-epoch cardiorespiratory embedding c_t,

$$\hat{y}^{\mathrm{apn}}_t = \sigma\big(W_2\,\mathrm{GELU}(W_1[h_t; c_t])\big). \tag{9}$$

The respiratory head receives the per-epoch cardiorespiratory embedding c_t directly in addition to the shared temporal representation. In the frozen-encoder configuration (Section 5), removing this bypass reduces respiratory

AUC from 0.782 to 0.776 ($p = 0.049$) and average precision from 0.419 to 0.411 ($p = 0.28$), while staging remains unchanged at 0.739 accuracy and $\kappa = 0.634$ ($p = 0.92$ for both). The ablation preserves the parameter count by zeroing. A control that permutes the same embeddings across epochs, preserving their distribution while disrupting temporal alignment, reaches 0.774 AUC. These results indicate a modest task-specific contribution from temporally aligned cardiorespiratory information at the respiratory head.

MM-Net holds 2.85 million trainable parameters. The sleep-adapted encoder that supplies the 200-dimensional neural embedding (5.8 million parameters) is fine-tuned in a separate stage per fold (Section 4.3) and run once to cache the embeddings, so it is not part of this count.

4.7. Training and Decoding

The two tasks are trained together with equal weight, $\mathcal{L} = \mathcal{L}^{\mathrm{stg}} + \lambda\,\mathcal{L}^{\mathrm{apn}}$ with $\lambda = 1$, combining a class-weighted staging cross-entropy and a positive-weighted respiratory cross-entropy,

$$\mathcal{L}^{\mathrm{stg}} = -\sum_{t}\sum_{k} w_k\, y^{\mathrm{stg}}_{t,k} \log \hat{y}^{\mathrm{stg}}_{t,k}, \tag{10}$$

$$\mathcal{L}^{\mathrm{apn}} = -\sum_{t} \left[\pi\, a_t \log \hat{y}^{\mathrm{apn}}_t + (1 - a_t)\log(1 - \hat{y}^{\mathrm{apn}}_t)\right]. \tag{11}$$

The final MM-Net uses uniform staging weights ($w_k = 1$), selected on validation patients together with the encoder (Table 13); the controlled analyses use the square root of inverse class frequency, $w_k \propto \sqrt{N/(K n_k)}$ for $K = 5$ classes with n_k examples, which corrects the scarce N1 and N3 stages. In both, the respiratory term is positive-weighted by $\pi = n_-/n_+$ to match the 16% event rate, which on this corpus is $\pi \approx 5.3$.

Both weightings shift the optimum of the loss, as follows. Against a true conditional rate p, the weighted binary term attains its minimum at $q^\star = \pi p/[\pi p + (1 - p)]$, which exceeds p whenever $\pi > 1$, so an epoch carrying a true event probability of 0.160 is driven towards 0.5; the staging weights displace the softmax by the same mechanism, to $q^\star_k \propto w_k p_k$. Neither head therefore emits a calibrated posterior, and a threshold of 0.5 on the respiratory output is not a fifty-per-cent event probability. Both displacements are monotone, so ranking is unaffected and the AUC and average precision we report are unchanged; the displacements affect only the interpretation of the outputs as probabilities, so the outputs are treated as scores and calibration is assessed separately (Section 5).

At inference the staging posteriors are smoothed by a Viterbi pass whose transition matrix A^{H} and prior are counted from the training labels [48],

$$\delta_t(j) = \max_i \left[\delta_{t-1}(i) + \log A^{\mathrm{H}}_{ij}\right] + \log \hat{y}^{\mathrm{stg}}_t(j), \tag{12}$$

recovered by back-tracking, which restores the natural persistence of sleep.

This is a transition-regularised discriminative decoder and not a generative hidden Markov model. A generative emission term is $\log P(\mathbf{x}_t \mid s_t = j)$, whereas $\hat{y}^{\mathrm{stg}}_t(j)$ is a class-weighted discriminative score computed from a bidirectional window; inserting it in the emission slot applies the stage prior a second time, once through A^{H} and once through the score itself. Dividing by the prior would not repair this, because the class weights and the recurrent context have already displaced the score from a calibrated posterior. The pass therefore functions as a transition prior on the output sequence. The respiratory head is left unsmoothed. Optimization uses AdamW with cosine annealing and early stopping on a held-out set of patients. Three seeds are trained per fold, and the final MM-Net is reported both as a single model (the mean over seeds) and as the three-seed average of the softmax outputs.

4.8. AHI Estimation

Because $\hat{y}^{\mathrm{apn}}_t$ scores event onsets, its sum over the epochs decoded as sleep estimates the number of events, and dividing by the decoded sleep time gives a rate,

$$\widehat{\mathrm{AHI}} = k \cdot \frac{\sum_{t:\,\hat{s}_t \neq \mathrm{W}} \hat{y}^{\mathrm{apn}}_t}{\frac{30}{3600}\,|\{t : \hat{s}_t \neq \mathrm{W}\}|}. \tag{13}$$

The positive weighting of the respiratory loss inflates the scores (Section 4.7), so a single scale k is fitted in each fold as the median ratio of annotated to raw estimate over the fold's validation patients and applied unchanged to its test patients. For transfer to another cohort, k is the median of the ten iSLEEPS fold values (0.459), and nothing is refitted on the external cohort. Severity follows the AASM bands (< 5, 5–15, 15–30 and ≥ 30 events/h).

5. Experiments

5.1. Protocol and Metrics

All results use ten-fold, patient-independent cross-validation, so no patient appears in both training and test, and within each training split a disjoint set of patients is held out for early stopping. Staging is scored by accuracy, macro-F1 $\frac{1}{K}\sum_k \mathrm{F1}_k$, and Cohen's κ. The respiratory task is scored by the area under the ROC curve (AUC) and average precision (AP), which suits the 16% event rate, and the AHI estimate by Spearman's ρ, the mean absolute error (MAE), severity-band accuracy and linearly weighted κ.

Two configurations of the same architecture are reported. The final MM-Net uses the sleep-adapted, cross-fitted and aligned encoder, eight cardiorespiratory channels, cross-attention fusion and uniform staging weights; it produces every headline result. The controlled analyses (respiratory bypass, fusion, modality ablation, joint objective, incomplete montage, permutation importance, stream representations, loss constants, calibration and nested selection) use the frozen-encoder configuration, which shares the streams, decoder and heads but reads a frozen LaBraM embedding of the EEG, seven cardiorespiratory channels, concatenation fusion and square-root class weights. Each of those analyses retrains dozens of variants, and adapting the encoder anew for every variant would multiply their cost by the four encoder trainings each fold requires. The frozen-encoder configuration reaches 0.739 accuracy (κ 0.634) and 0.782 respiratory AUC, and the quantities it is used to establish concern the streams, the objective and the representations that both configurations share.

Every number we report is a mean over the ten folds with its standard deviation. This standard deviation describes dispersion across folds and is not a confidence interval; because the folds share training patients, they are dependent. Because patients are approximately independent units, we also resample patients with replacement, 10,000 times, rebuilding the pooled epoch set and recomputing every metric. For the frozen-encoder configuration this gives 0.734 accuracy with a 95% interval of [0.717, 0.750], κ 0.628 [0.604, 0.651], respiratory AUC 0.778 [0.752, 0.802] and average precision 0.419 [0.355, 0.480]. These intervals are conditional on the fitted procedure: they reflect variation across the recorded patients but exclude retraining variability and the selection optimism quantified below. The folds are fixed across all configurations so that comparisons are paired. Fixed folds introduce a potential selection effect, quantified below: the frozen-encoder configuration was chosen by comparing forty-nine candidates on these same ten folds, which can introduce selection optimism into in-domain estimates. This optimism was measured by repeating the selection with the test patients held out from model selection (Section 5.2): five outer patient-independent folds, all forty-nine candidates compared on three inner folds within each outer training set, and the winner scored once on outer patients that took no part in choosing it. That procedure reaches 0.734 ± 0.009 accuracy against its 0.739, so the optimism is 0.006, about a third of the fold-to-fold standard deviation of 0.019, and it does not alter any comparison reported here. The external corpora in Section 5.19 took no part in selection either, so the transfer results reported there are not subject to this selection effect. The proposed model, its ablations and the retrained baselines all use the full ten; only the Sleep-EDF transfer row was run at five, as stated in the discussion of Table 5. Because performance estimates may vary across random seeds, we retrained the frozen-encoder configuration under three seeds on the same folds. Staging accuracy varies by 0.005 across seeds against 0.020 across folds, and respiratory AUC by 0.001 against 0.038: fold-to-fold spread is four times the seed spread for staging and about thirty-four times it for respiratory detection. The uncertainty in these results is therefore dominated by which patients fall in which fold, which is a property of a cohort this size. Every headline figure in this paper is the mean over all thirty fold-values, ten folds by three seeds. Paired tests, however, are computed on the ten fold-means: the three seeds within a fold train and test on the same patients, so they constitute repeated training runs on one patient partition, and treating them as thirty independent observations would inflate the effective sample size threefold and understate every p-value. Consequently, with ten paired folds the smallest two-sided Wilcoxon p is $2/2^{10} = 0.002$, reached when every fold moves the same way, so after Holm correction across a family of nine the floor is 0.018. Several effects below sit at that floor and differ in size by more than an order of magnitude; the test cannot separate them, and the effect sizes are reported beside every p for that reason. The comparisons of the respiratory bypass, the joint objective, the fusion type and the incomplete-montage subgroups are secondary analyses and are reported without multiplicity correction. The three seed means for respiratory AUC are 0.781, 0.783 and 0.781, so 0.782 is stable to the third decimal under reseeding.

5.2. Selection Inside the Training Data

The frozen-encoder configuration was chosen on the folds that score it; the encoder, fusion and class weights of the final MM-Net were chosen on validation patients only. To quantify the contribution of this selection, the entire

selection was repeated with the test patients held out from model selection: five outer patient-independent folds; inside each outer training set, three inner folds; all forty-nine candidates of the original search compared on those inner folds alone; and the inner winner retrained on the full outer training set and scored once on the outer patients. No outer patient influences the choice of the model that scores them, so the resulting estimate covers the complete procedure, including the search.

The nested procedure yields a comparable estimate, reaching 0.7336 ± 0.0089 accuracy across the five outer folds against 0.739 for the frozen-encoder configuration. The estimated selection optimism is therefore 0.006 accuracy, about a third of the fold-to-fold standard deviation and smaller than the performance differences underlying the principal comparisons. It is also close to the 0.004 suggested by the spread of the top ten candidates, indicating that the estimate available without a nested loop was of the correct order of magnitude.

Across the five outer folds, the neural configuration is selected consistently. The engineered descriptors, frozen pretrained encoder, BiLSTM, width, and regularisation are retained. The cardiorespiratory representation varies across folds: four select a randomly initialised time-series encoder over the raw signal, while one selects a random projection of the fourteen engineered descriptors; none selects the learned convolutional encoder.

Together with the representation experiments in Section 5.15, this indicates that richer cardiorespiratory representations are useful and that no single encoder architecture is consistently preferred at this cohort size. The selected configurations remain within 0.006 staging accuracy of the frozen-encoder configuration.

Because each outer model is trained jointly, the nested procedure also provides a selection-free estimate for the respiratory output. Across the five outer folds, respiratory AUC is 0.774 ± 0.047 and average precision is 0.406 ± 0.077, corresponding to differences of 0.008 AUC and 0.013 average precision relative to the frozen-encoder configuration. These differences are similar in scale to the staging selection effect and are small relative to the respiratory fold-to-fold standard deviation of 0.038. The inner loop selects models using staging accuracy. The four outer folds selecting the randomly initialised time-series encoder average 0.793 respiratory AUC; the remaining fold, which selected the random projection of the fourteen descriptors, yields 0.698. The nested estimate therefore remains above the 0.698 obtained with the fourteen engineered descriptors in the same configuration (Table 12).

The nested estimate uses five outer folds and a single training seed and is therefore less precise than the ten-fold, three-seed headline evaluation.

5.3. Implementation and Hyperparameters

The model is implemented in PyTorch and trained on single NVIDIA RTX 2060 (6 GB) and RTX 3060 Ti (8 GB) GPUs under Windows. Every run is seeded, so each is individually reproducible; headline figures average three seeds (42, 1, 7) over the ten folds, giving the thirty fold-values used throughout. Where a result is derived from stored predictions without retraining, it comes from the seed-42 run, and this is stated where it applies. Table 3 lists the training configuration, which is identical across all folds and every ablation, so that reported differences reflect the model configuration. The six reimplemented baselines are trained at the optimiser, batch size and schedule of their own papers, because imposing our settings could disadvantage architectures tuned for other settings.

5.4. Compared Methods

We compare the model against deep networks evaluated on the same folds: convolutional and recurrent networks trained from scratch (DeepSleepNet, AttnSleep, a four-channel CNN–BiLSTM), six further published architectures we reimplemented for this comparison (U-Time, a fully convolutional encoder–decoder; TinySleepNet, a compact convolutional–recurrent model; SleepTransformer, a purely attentional one; and 1D-ResNet-SE-LSTM, Micro SleepNet and AT-BiLSTM), the released SE-ResNet-18 implementation from [22], a network transferred from Sleep-EDF, and a raw-signal convolutional network on all 14 channels, alongside the published single-channel baselines for the corpus [21]. For the respiratory task, a desaturation rule, logistic regression and gradient boosting read the 14 engineered cardiorespiratory descriptors, and two raw-signal detectors read the waveforms (Section 5.10). On ISRUC-I, the reimplemented staging baselines are retrained on MM-Net's folds (Section 5.19). All numbers are patient-independent. All use ten folds except the Sleep-EDF transfer row, which uses five; hidden-Markov decoding is applied where the corresponding model provides it.

5.5. Staging Benchmark and Comparison

The row marked ‡ in Table 5 reports the CNN-ResNet18 result published in [22] under that study's $N = 100$ protocol, which includes the duplicate recording excluded here (Section 3). The same study reports 0.747 accuracy,

Table 3
Training configuration of the final MM-Net, shared across all folds and cohorts. The controlled analyses use the same settings with a frozen encoder, concatenation and square-root class weights. Each reimplemented baseline is trained at its own published settings.

Setting	Value
Framework / hardware	PyTorch, NVIDIA RTX 2060 / 3060 Ti
Sleep-adapted encoder (per fold)	
Encoder	LaBraM-Base, 4 EEG + 2 EOG
Encoder LR (head)	3×10^{-5} (10^{-3}), one-cycle
Passes, batch, weight decay	3, 64, 0.05
Cross-fitting, alignment	$K = 3$, ridge $\lambda = 0.01$
MM-Net	
Optimizer	AdamW
Learning rate	3×10^{-4}
Weight decay	1×10^{-4}
LR schedule	cosine annealing
Sequence length L	20 epochs (10 min)
Batch size	32
Max epochs / patience	45 / 8
Gradient clipping	5.0
Fusion	cross-attention, 4 heads
Staging class weights	uniform
Respiratory pos. weight	$\pi = n_-/n_+$
Loss balance λ	1.0
Staging post-processing	HMM Viterbi smoothing
Cross-validation	10-fold, patient-independent
Random seeds	42, 1, 7

Table 4
Reimplementation checks on Sleep-EDF for the models marked † in Table 5. Published and reproduced values are compared under their respective reported protocols; U-Time is compared using macro-F1, while the remaining rows use accuracy.

Model	Published	Reproduced	Δ
U-Time [23]§	0.79	0.660 ± 0.054	−0.130
TinySleepNet [8]	0.854	0.824 ± 0.045	−0.030
SleepTransformer [25]	0.814	0.823 ± 0.033	+0.009
1D-ResNet-SE-LSTM [49]	0.864	0.832 ± 0.036	−0.032
Micro SleepNet [50]	0.828	0.800 ± 0.041	−0.028
AT-BiLSTM [51]	0.838	0.835 ± 0.046	−0.003

0.677 macro-F1, and $\kappa = 0.64$ for an SE-ResNet-18 + BiLSTM model. To obtain a comparison under the common protocol used in this study, we retrained the authors' released implementation on our 99-patient folds using its released training and decoding procedure. The rerun reaches 0.696 ± 0.038 accuracy, 0.606 ± 0.032 macro-F1, and $\kappa = 0.572 \pm 0.046$ (single seed), compared with 0.763, 0.664, and 0.656, respectively, for MM-Net. The published and same-fold results are kept separate because the original evaluation differs in patient set, model selection, and epoch-scoring procedure; Table 5 therefore uses the same-fold rerun for quantitative comparison while retaining the published value as literature context. Each reimplementation is checked against the figure its own paper publishes on Sleep-EDF (Table 4); those figures follow each paper's own protocol, which differs in subset and split, so they are consistency checks. U-Time is the only row compared in macro-F1 because its paper does not report Sleep-EDF accuracy; its published value uses 39 recordings under twenty-fold cross-validation, whereas our check uses 40 recordings under ten folds. Table 4 therefore serves as an implementation consistency check across the reported protocols.

Table 5 is organised in four blocks: architectures retrained on this cohort, architectures trained on healthy sleep and applied to it zero-shot, the sleep-adapted encoder used on its own, and MM-Net. The Healthy column gives each

Table 5
Patient-independent sleep-staging comparison on iSLEEPS. Values are mean ± SD where fold-level results are available. † denotes reimplemented models, ‡ a literature-reported result under its original protocol, and * models ported from released code. Bold indicates the best result among models evaluated under our protocol.

			Healthy	iSLEEPS (stroke)		
Model	Input	Params ↓	Acc	Acc ↑	Macro-F1 ↑	κ ↑
Comparative models evaluated on iSLEEPS						
LSTM, our reimplementation†	1 EEG + EOG	1.16 M	0.820 ± 0.062	0.720 ± 0.032	0.626 ± 0.028	0.599 ± 0.043
CNN-ResNet18 [22]‡	1 EEG	∼11 M	—	0.617	0.544	0.480
SE-ResNet-18 + BiLSTM [22]*	1 EEG	11.32 M	0.829 ± 0.056	0.696 ± 0.038	0.606 ± 0.032	0.572 ± 0.046
DeepSleepNet [5]	4 EEG	3.72 M	0.823 ± 0.050	0.669 ± 0.031	0.624 ± 0.026	0.558 ± 0.035
U-Time [23]†	1 EEG	1.07 M	0.791 ± 0.062	0.646 ± 0.064	0.524 ± 0.057	0.506 ± 0.082
AttnSleep [7]	1 EEG	∼0.5 M	0.833 ± 0.042	0.716 ± 0.042	0.600 ± 0.043	0.593 ± 0.058
TinySleepNet [8]†	1 EEG	1.45 M	0.824 ± 0.045	0.676 ± 0.050	0.587 ± 0.034	0.548 ± 0.062
SleepTransformer [25]†	1 EEG	2.79 M	0.823 ± 0.033	0.683 ± 0.026	0.602 ± 0.030	0.555 ± 0.032
1D-ResNet-SE-LSTM [49]†	1 EEG	0.65 M	0.832 ± 0.036	0.708 ± 0.038	0.631 ± 0.033	0.592 ± 0.050
Micro SleepNet [50]†	1 EEG	0.03 M	0.800 ± 0.041	0.647 ± 0.049	0.558 ± 0.028	0.510 ± 0.055
AT-BiLSTM [51]†	1 EEG	0.92 M	0.835 ± 0.046	0.692 ± 0.040	0.612 ± 0.030	0.570 ± 0.051
CNN + BiLSTM	4 EEG	∼0.6 M	0.806 ± 0.041	0.670 ± 0.052	0.618 ± 0.044	0.553 ± 0.062
Sleep-EDF transfer	4 EEG	∼0.6 M	—	0.623 ± 0.020	0.582	0.488
Raw multimodal CNN	14 ch	0.95 M	—	0.677 ± 0.035	0.633 ± 0.025	0.563 ± 0.041
Huttunen et al. [13]*	4 ch	3.80 M	—	0.599 ± 0.060	0.495 ± 0.067	0.440 ± 0.078
Trained on healthy sleep, applied zero-shot						
CNN	4 EEG	0.49 M	0.806 ± 0.001	0.307 ± 0.016	0.223 ± 0.013	0.102 ± 0.010
CNN + BiLSTM (CRNN)	4 EEG	1.15 M	0.828 ± 0.005	0.312 ± 0.012	0.253 ± 0.006	0.127 ± 0.022
StagingSeqNet	4 EEG	1.08 M	0.855 ± 0.001	0.306 ± 0.009	0.230 ± 0.014	0.091 ± 0.015
Sleep-adapted encoder alone (one epoch, no temporal model)						
LaBraM, frozen + logistic regression	4 EEG	5.8 M	—	0.647	—	—
LaBraM, sleep-adapted	4 EEG + 2 EOG	5.8 M	—	0.732	—	—
Proposed multi-task model						
MM-Net, single model	7 neural + 8 cardio	2.85 M	0.864 ± 0.030	0.756 ± 0.013	0.661 ± 0.019	0.647 ± 0.015
MM-Net, three-seed	7 neural + 8 cardio	2.85 M	**0.864 ± 0.030**	**0.763 ± 0.014**	**0.664 ± 0.020**	**0.656 ± 0.016**

model's subject-independent accuracy on Sleep-EDF; MM-Net's healthy-sleep macro-F1 and κ are 0.791 and 0.810. Dispersion is reported wherever it is defined: five folds for the Sleep-EDF transfer row; ten fold-values (seed 42) for DeepSleepNet, the CNN + BiLSTM, the raw multimodal CNN and the SE-ResNet-18 model; thirty fold-values, ten folds by three seeds, for AttnSleep, the reimplemented literature models and the proposed rows. CNN-ResNet18 is quoted from [22] and was not re-run here, so its protocol is that paper's and it carries no interval. That paper's LSTM appears only as our reimplementation of it on the folds used throughout, because its published figure was never measured on these folds. In the zero-shot block the interval is across pretraining seeds and not folds, because that evaluation is a single pass of a finished checkpoint over all 99 patients. MM-Net reaches 0.864 on healthy sleep, the highest healthy accuracy in the table. Sleep-EDF carries none of the cardiorespiratory channels this model reads, its Cassette recordings providing an oronasal airflow trace at 1 Hz but no ECG, effort belts or oximetry and no scored respiratory events, so MM-Net's healthy-sleep figure exercises the neural stream alone.

Across the evaluated models retrained on iSLEEPS, the deep architectures reach 0.60–0.72 staging accuracy, while MM-Net reaches 0.756 as a single model and 0.763 as a three-seed average, with κ 0.656. The same-fold SE-ResNet-18 + BiLSTM rerun reaches 0.696, and the corpus LSTM reaches 0.720 ± 0.032. The sleep-adapted encoder used on its own stages single epochs at 0.732, already above every retrained architecture; within MM-Net, together with the engineered descriptors, the cardiorespiratory stream and the temporal decoder, it yields 0.756, a further 0.024 in eight folds of ten ($p = 0.014$). MM-Net additionally provides respiratory-event prediction and AHI estimation within the same model. The raw multimodal CNN also produces two outputs but stages at 0.677.

The closest task-matched baseline is Huttunen et al., whose model jointly scores sleep stages and respiratory events. Ported from the authors' released implementation and evaluated on the same iSLEEPS folds, it reaches 0.599 staging

accuracy, against 0.756 for the single MM-Net model, which is ahead in all ten folds (Wilcoxon $p = 0.002$). On ISRUC-I, where both models are trained and evaluated on the same respiratory labels, MM-Net is also ahead on respiratory detection (0.889 against 0.863 AUC; Section 5.19).

The comparison is specific to the adapted iSLEEPS implementation. The port uses four of the five signals in the fullest original configuration, excludes nasal pressure, maps the original per-second three-class respiratory output to the 30-s binary target used here, and uses a single EEG derivation. These differences define the scope of the same-cohort comparison.

A representative held-out night is shown in Figure 3: the predicted hypnogram follows the reference through the night's cycles, and the stage-probability ribbon shows the epochs with high and low posterior confidence.

Paired comparisons use the same patient folds across the thirteen baselines retrained in this study with fold-level results. As a single model, MM-Net is higher in accuracy by +0.036 (corpus LSTM) to +0.157 (Huttunen et al.) and is higher in nine or ten of the ten folds for every comparison; all thirteen differences remain significant after Holm correction (Holm $p \leq 0.025$). The three-seed MM-Net is ahead of each of the eight reimplemented and ported architectures in all ten folds (Holm $p = 0.016$, the minimum attainable for eight comparisons). Differences in macro-F1 and κ are larger than those observed for accuracy, consistent with their reduced dependence on performance in the majority classes.

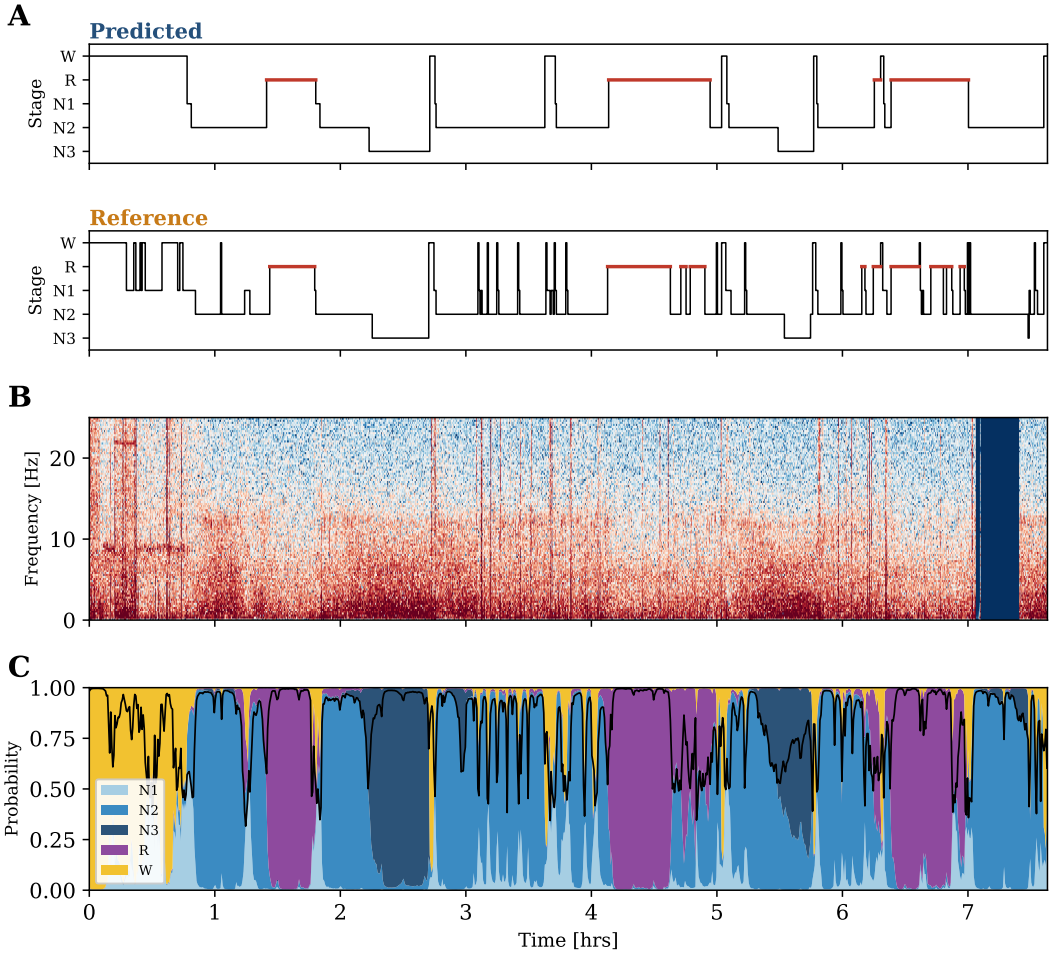


Figure 3: An illustrative held-out night (patient SN90, three-seed MM-Net, staging accuracy 0.83). **(A)** Predicted and reference hypnograms, REM highlighted. **(B)** EEG spectrogram, C4 derivation, 0–25 Hz. **(C)** Per-epoch stage-probability ribbon; the black line is the maximum posterior.

Table 6
Per-class staging F1 (Wake, N1, N2, N3, REM), ten-fold patient-independent, at the hidden-Markov operating point (MM-Net: three-seed, pooled test epochs). Best per column in **bold**.

Model	W ↑	N1 ↑	N2 ↑	N3 ↑	R ↑
Raw multimodal CNN	0.74	**0.37**	0.73	0.56	0.67
MM-Net	**0.81**	0.20	**0.82**	**0.72**	**0.79**

5.6. Per-class Staging Performance

Table 6 breaks staging performance down by stage. MM-Net recovers Wake, N2 and REM at F1 0.79–0.82 and N3 at 0.72; REM, whose ocular signature the sleep-adapted encoder receives, reaches 0.79. N1, the transient stage between wake and sleep, has the lowest F1 for every model on this cohort, and the raw multimodal CNN attains the highest N1 F1, whereas MM-Net performs more strongly across the remaining stages. The row-normalised confusion matrix (Figure 4) shows the distribution of errors: N1 is absorbed mainly into Wake and N2, and part of N3 is read as N2. In the frozen-encoder configuration, comparing the neural-only and full models shows little change in staging after adding the cardiorespiratory stream: overall accuracy changes from 0.741 to 0.739, N1 F1 increases by 0.019 ($p = 0.11$), and no per-stage difference reaches statistical significance ($p \geq 0.15$).

These results are consistent with the modality-ablation analysis in Section 5.7: staging depends primarily on the neural stream, whereas the cardiorespiratory stream supplies respiratory-event information and leaves staging performance essentially unchanged.

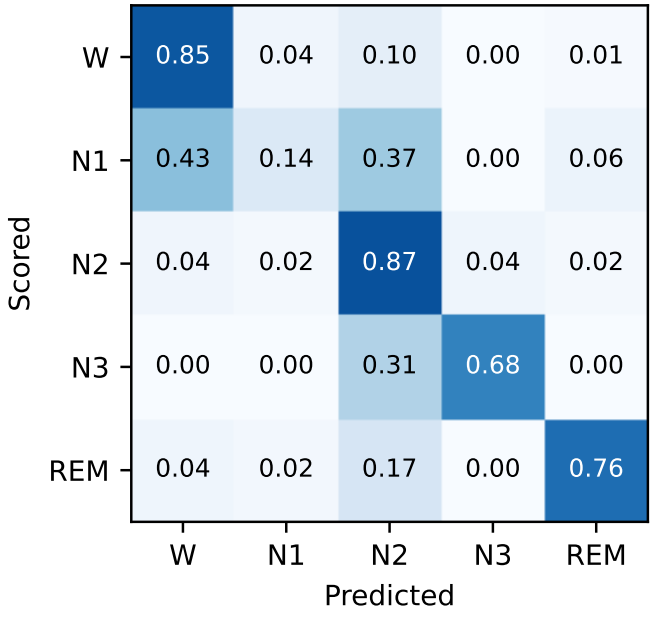


Figure 4: Row-normalised confusion matrix for MM-Net (three-seed, pooled ten-fold test epochs, hidden-Markov operating point). Errors concentrate on N1, which is absorbed into Wake and N2.

5.7. Modality-Ablation Grid

This ablation is also our attribution method. Because the trainable network receives engineered descriptors and precomputed embeddings, and no raw spectrogram or EEG trace, a gradient heat-map or Grad-CAM would have no faithful spatial support on the neural side, which is where a staging attribution would be read; we therefore attribute each output by *interventional* ablation, deleting a physiological channel and measuring the change. This is a falsifiable test of what each modality contributes. The analysis uses the frozen-encoder configuration (Section 5), which stages

sleep at 0.739 accuracy (κ 0.634) and detects respiratory events at 0.782 AUC (0.419 average precision). To test whether each modality carries the task to which its physiology is most closely related, we remove each modality in turn and keep the rest, on the same folds, with three seeds giving thirty fold-values per row (Table 8). The results show a task-specific separation between streams. Within the cardiorespiratory stream, the pattern differs from that obtained with the engineered features.

Removing the whole cardiorespiratory stream drops respiratory AUC from 0.782 to 0.654, the largest effect in the ablation grid, and leaves staging untouched (0.741, n.s.). Removing the neural stream does the reverse: staging falls from 0.739 to 0.680 while respiratory detection holds. No neural ablation significantly moves respiratory AUC and no cardiorespiratory ablation significantly moves staging, so each stream serves the outcome its physiology is closest to.

Within the cardiorespiratory stream, however, only respiratory effort is individually necessary (-0.038 AUC, Holm $p = 0.018$). Removing SpO_2, pulse/HRV, ECG or airflow individually costs at most 0.005 AUC and none survives correction; SpO_2, the central input of the engineered-feature configuration, is among the least necessary channels here (-0.002, $p = 1.00$). The five individual removals sum to -0.046 AUC against -0.128 for removing the stream as a whole, a factor of 2.8. The learned encoder therefore distributes respiratory information redundantly across channels: each is individually dispensable because the others overlap with it, but the channels jointly are essential. That redundancy is clinically favourable, since twelve of the ninety-nine recordings are missing at least one cardiorespiratory channel, and it is physiologically coherent: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event as it happens, whereas desaturation is a lagged consequence of it.

5.8. What the Joint Objective Contributes

As for the ablation grid, this comparison uses the frozen-encoder configuration. The ablation grid establishes which physiology carries which output. It does not establish whether the two outputs should share a network at all: a model producing two outputs demonstrates integration, and integration alone does not demonstrate benefit. The appropriate control is a pair of single-task models matched to the joint model in every other respect. The architecture, the ten folds, the three seeds, the optimiser, the budget and the cardiorespiratory bypass are those of the full model; only the surviving term of $\mathcal{L} = \mathcal{L}^{\mathrm{stg}} + \lambda\mathcal{L}^{\mathrm{apn}}$ differs, so the staging-only model still builds the cardiorespiratory stream and still feeds the respiratory head but never receives a respiratory gradient.

Joint training improves on training either task alone (Table 7). Staging rises from 0.733 to 0.739 accuracy, in nine folds of ten (Wilcoxon on the ten fold-means, $p = 0.014$), and κ from 0.625 to 0.634 on the same nine. Respiratory AUC rises from 0.774 to 0.782 in seven folds of ten ($p = 0.049$). Average precision moves $+0.005$ and does not separate ($p = 0.63$), and is reported as a null result.

These gains support a narrow interpretation. The gains are small: 0.006 accuracy against a fold-to-fold SD of 0.019, and 0.007 AUC against 0.038, approximately one third and one fifth of the fold-to-fold variation. Their significance derives from consistency across folds, with nine and seven of ten folds moving in the same direction, which the paired test evaluates. The supported conclusion is that adding the respiratory task does not reduce staging performance and yields a small improvement, so the respiratory output is obtained without a loss of staging accuracy. We do not claim multi-task learning as a source of large gains here.

Two diagnostic cells verify the comparison. The respiratory-only model stages at 0.303 accuracy and κ 0.039, and the staging-only model detects at 0.495 AUC. Both are at chance for their untrained head, which confirms that the switch isolated the loss and that the other objective was inactive.

5.9. Recordings With an Incomplete Montage

In the frozen-encoder configuration, those twelve recordings are scored with the absent channels zero-filled instead of being excluded, which raises the question of whether the network relies on missingness patterns instead of physiology. The evidence indicates that it does not. Removing all twelve from the pooled respiratory evaluation moves AUC from 0.777 to 0.775 and moves average precision the other way, from 0.419 to 0.439, so the respiratory result does not depend on these recordings. Predicted per-patient burden is independent of whether a montage was complete ($p = 0.91$), so missingness provides no usable shortcut.

Staging is lower on those twelve, 0.670 against 0.743 ($p = 0.017$), which requires a control comparison. The neural-only configuration receives no cardiorespiratory channel at all and so has nothing zero-filled, yet it shows the same gap in the same direction, 0.689 against 0.744 ($p = 0.040$). Most of the difference is therefore attributable to the recordings themselves, which are also harder for a model that does not receive the missing channels. On those twelve

Table 7
Joint multi-task training compared with matched single-task models using the same architecture, folds, seeds, and training budget (frozen-encoder configuration). The lower rows report the untrained head as a control for loss isolation.

Trained on	Metric	Single	Joint	p
Staging only	Accuracy	0.733	**0.739**	**0.014**
	κ	0.625	**0.634**	**0.014**
Respiratory only	AUC	0.774	**0.782**	**0.049**
	AP	0.414	0.419	0.63
Untrained heads, reported as the check that the switch worked				
Respiratory only	Accuracy	0.303	—	—
Staging only	AUC	0.495	—	—

Table 8
Modality-ablation grid (leave-one-out, frozen-encoder configuration), ten-fold patient-independent, three seeds. Each row removes one modality and keeps the rest; cells are means over thirty fold-values. p is Wilcoxon on the ten fold-means, Holm-corrected across the nine ablations within each metric and **bold** where it clears 0.05; the floor at this sample size is 0.018.

Modality removed	Staging acc ↑	Holm p	Respiratory AUC ↑	Holm p
none (full model)	0.739 ± 0.020	—	0.782 ± 0.038	—
EEG (neural)	0.680	**0.018**	0.774	0.92
EOG (ocular)	0.731	0.08	0.777	1.00
EMG (muscle)	0.735	1.00	0.778	1.00
SpO_2	0.742	1.00	0.780	1.00
respiratory effort	0.741	1.00	0.744	**0.018**
pulse / HRV	0.741	1.00	0.785	1.00
ECG	0.737	1.00	0.777	1.00
airflow	0.740	1.00	0.778	1.00
all cardiorespiratory	0.741	1.00	0.654	**0.018**

the full model sits 0.019 below neural-only, which at $n = 12$ is not distinguishable from noise (Wilcoxon $p = 0.18$), so a small cost can be bounded but not excluded. On the eighty-seven complete recordings the two agree to within 0.002.

A zero-filled channel can resemble a flat channel after per-patient standardisation, so we also retrained two variants with explicit missing-input information on the same folds using seed 42. The first appends seven per-recording channel-availability flags to the pooled cardiorespiratory representation. The second combines these flags with channel dropout, removing each available channel and its flag during training with probability 0.15. Under the same seed, the reference configuration reaches 0.734 accuracy and 0.779 respiratory AUC, with AUCs of 0.779 on the 87 complete recordings and 0.791 on the 12 incomplete recordings. Availability flags yield 0.739 accuracy and 0.782 pooled AUC, with subgroup AUCs of 0.776 and 0.835; flags with channel dropout yield 0.738, 0.779, 0.770, and 0.826, respectively. Pooled performance is therefore essentially unchanged. Both explicit missingness treatments increase AUC on the incomplete-recording subgroup by about 0.04; with twelve patients and one seed, this subgroup difference is treated as suggestive. In this configuration, attention-based and concatenation fusion yield similar staging and detection performance (Section 4), so the fusion mechanism has little effect on these results.

5.10. Physiological Validity of the Respiratory Read-out

Because the events were scored from airflow, predominant reliance on that channel could produce a circular association. In the frozen-encoder configuration, removing the airflow features leaves detection essentially unchanged, at 0.778 AUC (Table 8), within one fold standard deviation of the full 0.782. After retraining, the remaining scoring-relevant channels recover nearly all respiratory discrimination. Respiratory effort provides the strongest individual contribution; the smaller effects of the other single-channel removals indicate that the learned encoder represents them redundantly. This indicates that detection is supported by physiological correlates of disordered breathing. This pattern

is consistent with physiology: the effort belts register the paradoxical thoraco-abdominal motion of an obstructive event while it is happening, whereas desaturation follows it by several breaths.

5.11. Respiratory Baselines and Per-event-type Detection

The respiratory task is compared with established detectors on the same folds and the same epoch labels (Table 9). The classical baselines read the 14 engineered cardiorespiratory descriptors; MM-Net reads the raw signal through its learned encoder (in the frozen-encoder configuration its own score on the 14 descriptors is 0.698, Table 12). A clinical desaturation rule reaches 0.595 AUC, logistic regression 0.603, and gradient boosting 0.673; a random classifier scores 0.160 average precision, the event prevalence. MM-Net reaches 0.825 AUC and 0.487 average precision, 0.152 AUC and 0.197 average precision above gradient boosting. Detection also separates by event type: the model scores 0.810 AUC on hypopneas (the mild, dominant 82% of event epochs), 0.898 on obstructive apneas, and 0.946 on central apneas. It is strongest on the most severe events, and the pooled AUC is dominated by the more numerous hypopneas.

To compare the proposed representation against end-to-end waveform detectors, we added two raw-signal respiratory baselines. Each uses a per-epoch convolutional encoder over the 25 Hz waveform followed by a bidirectional LSTM across 20-epoch windows and is trained with the same epoch labels, folds, early-stopping rule, and positive-class weighting (n_-/n_+). The first reads summed respiratory effort alone, matching the input configuration of Nassi *et al.* [40]; the second reads all seven raw cardiorespiratory channels. With a single seed, the effort-only model reaches 0.602 ± 0.057 AUC and 0.252 average precision, while the seven-channel model reaches 0.733 ± 0.036 AUC and 0.366 average precision (Table 9). MM-Net reaches 0.825 AUC and 0.487 average precision using these channels and nasal pressure together with the neural stream and shared temporal context.

Figure 5 shows the operating characteristics behind these numbers, computed from the pooled held-out predictions of all ten folds. The precision–recall panel is the more informative of the two: respiratory events occupy 16% of epochs, and under this class imbalance a ROC curve can overstate detector performance. Read at fixed sensitivity, the trade-off is explicit: at 0.80 sensitivity MM-Net holds 0.70 specificity, and at 0.90 sensitivity 0.53. Pooling every fold's epochs into a single curve is not the same estimator as averaging the ten per-fold scores: pooling weights folds by epoch count and mixes their score scales, giving a pooled AUC of 0.828 against the cross-validated 0.825 ± 0.035 reported in Table 9; at 0.003 the difference is a tenth of one fold standard deviation.

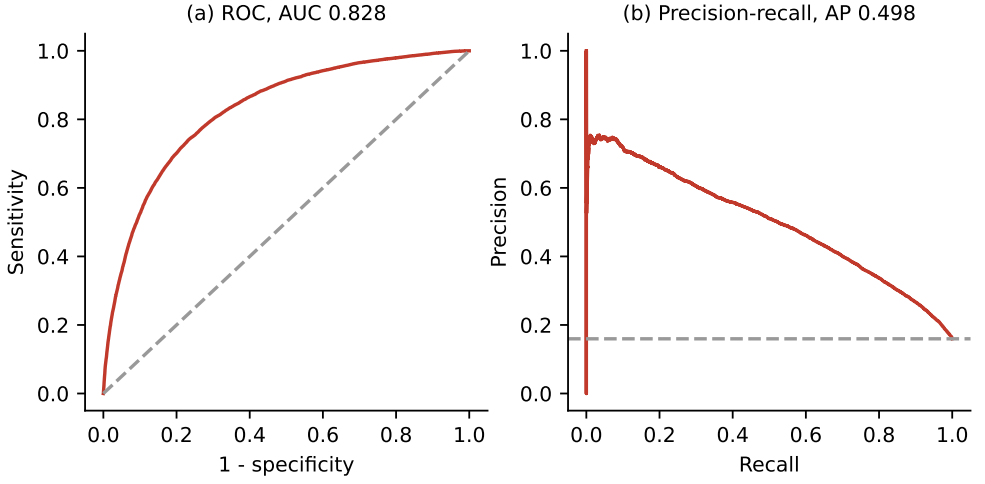


Figure 5: Respiratory-event detection by MM-Net (three-seed), pooled over the ten patient-independent test folds. **(a)** ROC curve. **(b)** Precision–recall; the dashed line is the 0.160 event prevalence, the no-skill floor.

5.12. AHI Estimation and Clinical Severity

MM-Net's AHI estimate (Section 4.8) tracks the annotated index across the full severity range: Spearman $\rho = 0.765$ over the 99 test patients, a mean absolute error of 10.4 events/h, and 54% of patients placed in the correct AASM severity band (weighted κ 0.45; Table 10, Figure 6). Two references place this result. The clinical AHI recorded with the corpus

Table 9
Respiratory-event detection on iSLEEPS. (A) Comparison with classical and raw-signal baselines on the same folds and (B) MM-Net AUC by respiratory-event type. Bold denotes the best value in each comparison.

(a) Detection method

Method	AUC ↑	AP ↑
Desaturation rule	0.595	0.207
Logistic regression	0.603	0.211
Gradient boosting	0.673	0.290
Raw effort belt[§] [40]	0.602	0.252
Raw seven channels[§]	0.733	0.366
MM-Net, single model	0.815	0.471
MM-Net, three-seed	**0.825**	**0.487**

(b) MM-Net AUC by event type

Event type	Positive epochs	AUC ↑
Hypopnea	11,921	0.810
Obstructive	1,722	0.898
Central	832	**0.946**

Table 10
AHI estimation on iSLEEPS (99 test patients, stages decoded by MM-Net). Severity: AASM bands; κ_w linearly weighted. Last column: Spearman ρ against the clinical AHI recorded with the corpus. The last row gives the agreement of the clinical AHI with the annotated AHI.

Estimator	ρ ↑	MAE ↓	Sev. acc ↑	κ_w ↑	ρ_{clin} ↑
Event detector	0.669	11.7	0.45	0.44	0.569
MM-Net	**0.765**	10.4	0.54	0.45	**0.633**
MM-Net + event detector	0.742	**9.7**	**0.55**	**0.51**	0.625
Clinical vs. annotated AHI	0.687				

agrees with the annotated index at $\rho = 0.687$, so MM-Net's estimate is closer to the annotations than the clinical index itself; against that clinical index MM-Net reaches $\rho = 0.633$. In the expected-count formulation, summing the onset scores over decoded sleep counts closely spaced events separately, whereas thresholding the scores into runs would merge them, and normalising by decoded sleep time matches the denominator of the clinical index.

We also trained a dedicated event detector, a one-dimensional U-Net with a recurrent bottleneck operating at one-second resolution on the raw cardiorespiratory signals, with its counting rule chosen on validation patients. On its own it reaches $\rho = 0.669$; averaged with MM-Net it lowers the error to 9.7 events/h and raises severity agreement to 55% (weighted κ 0.51), and MM-Net alone yields the highest correlation. The AHI is therefore available from the staging model itself, without a separate detector.

Staging remains stable across sleep-disordered-breathing severity: accuracy is 0.784 in patients with a normal clinical AHI, 0.754, 0.764 and 0.758 in the mild, moderate and severe groups, with no significant ordered trend (Jonckheere–Terpstra $p = 0.20$; Table 11). The model therefore stages the most severely affected patients nearly as well as those without sleep-disordered breathing, the group in which both outputs are most clinically relevant.

Table 11
Sleep-staging accuracy of MM-Net across AASM severity groups of the clinical AHI in iSLEEPS. Values are mean ± SD across patients; the ordered-group trend is not significant (Jonckheere–Terpstra $p = 0.20$).

AHI band	AHI (events/h)	Patients	Staging acc ↑
Normal	< 5	15	0.784 ± 0.057
Mild	5–15	23	0.754 ± 0.092
Moderate	15–30	23	0.764 ± 0.083
Severe	≥ 30	38	0.758 ± 0.078

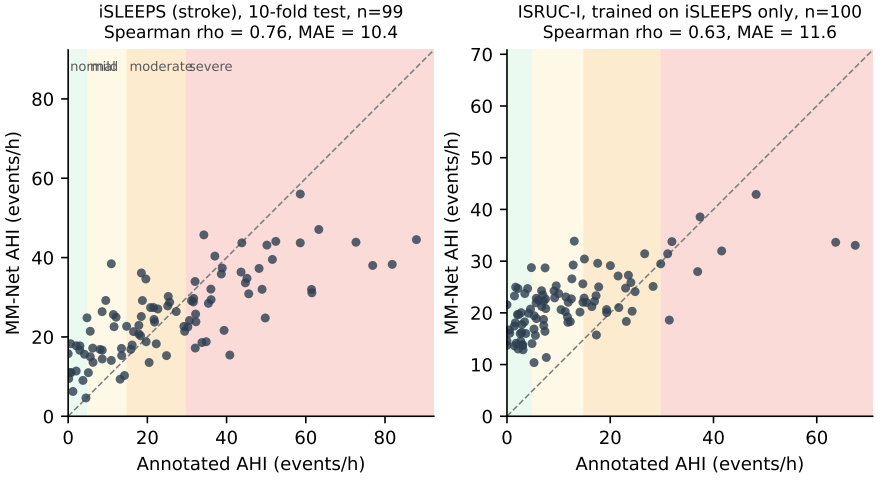


Figure 6: MM-Net AHI estimate against the annotated AHI. Left: iSLEEPS, ten-fold test patients. Right: ISRUC-I, MM-Net trained on iSLEEPS only and applied without retraining (Section 5.19). Shading marks the AASM severity bands; the dashed line is identity.

5.13. Permutation Importance: a Second, Independent Attribution

The ablation grid establishes what the model *needs*: remove a modality, retrain, and observe which output degrades. A complementary question is what a *trained* model actually uses. We therefore permute each modality's features across epochs at inference time in the frozen-encoder configuration, so the channel is still present but no longer aligned to the epoch it describes, and measure the drop, averaged over three shuffles and the ten fold models.

The two methods can disagree, because a model may need a channel during training yet weight it lightly afterwards. On staging they do not: permuting the neural features costs 0.284 staging accuracy against 0.030 for the ocular and 0.006 for the muscle channel, the same ordering the ablation gives, and the rank correlation between the two attributions is $\rho = 0.833$ ($p = 0.010$).

On the respiratory head the agreement is narrower, and we report the two attributions separately. Both methods place respiratory effort first by a wide margin: 0.038 AUC under retraining, 0.071 under permutation, in each case substantially above the second-ranked channel. The remaining four cardiorespiratory channels, however, have retraining effects that are not distinguishable from zero, so their rank order is not informative and the correlation over the five is correspondingly uninformative ($\rho = 0.381$, $p = 0.35$).

Permutation effects are largest for respiratory effort (0.071 AUC), airflow (0.027), and SpO_2 (0.016), followed by pulse/HRV (0.012), with the remaining channels below 0.006. The magnitude ordering therefore identifies respiratory effort as the dominant input used by the fitted respiratory model. Retraining and permutation measure complementary properties. Retraining after channel removal measures whether the remaining inputs can compensate for the missing signal, whereas permutation measures reliance of the fitted model on the aligned input. SpO_2, for example, produces a 0.002 AUC change after retraining without it but a 0.016 decrease when shuffled at inference. Both analyses identify respiratory effort as the strongest individual cardiorespiratory contributor.

5.14. Healthy-Trained Models Applied to the Injured Brain

To examine cross-dataset transfer without adaptation, we trained three staging architectures on Sleep-EDF: a four-channel CNN, a CNN with a BiLSTM over epochs, and StagingSeqNet. Their Sleep-EDF accuracies range from 0.806 to 0.855 with $\kappa = 0.73$–0.81. Applied zero-shot to the 99 iSLEEPS patients, the same checkpoints reach 0.306–0.312 accuracy with $\kappa = 0.09$–0.13. Predictions shift strongly toward Wake, which accounts for 62–79% of predicted epochs, while N2 accounts for only 1–5% despite representing 42.3% of the iSLEEPS epochs.

A held-out Sleep-EDF implementation check using the same preprocessing reaches 0.8682 accuracy (0.8667 ± 0.0415 per recording), confirming that the healthy-sleep performance is retained through the evaluation pipeline. The zero-shot result must nevertheless be interpreted under both population and montage shift: Sleep-EDF uses Fpz-Cz and Pz-Oz, whereas iSLEEPS uses C4 and O2. The transfer experiment therefore establishes a substantial dataset shift but does not isolate stroke as its sole source.

MM-Net reaches 0.864 ± 0.030 accuracy with $\kappa = 0.810$ under subject-independent evaluation on Sleep-EDF, compared with 0.763 on iSLEEPS. Across the retrained staging baselines, the corresponding healthy-to-iSLEEPS decreases range from approximately 0.12 to 0.15. These cross-system differences are descriptive because channel configurations and representations differ across models, but they consistently show that staging performance depends strongly on the target dataset and population.

Sleep-EDF folds are split at the subject level because individual participants contribute two recording nights, preventing recordings from the same subject from appearing in both training and test folds.

5.15. Stream Representations

The representation experiments use the frozen-encoder configuration: engineered features together with a frozen self-supervised embedding for the neural stream and a learned convolutional encoder over the raw cardiorespiratory signal. Both representation choices were evaluated experimentally before selection. Each family below holds every other hyperparameter fixed and varies one factor, over the same three seeds and ten patient-independent folds, with Holm correction within the family and every test on the ten fold-means.

The cardiorespiratory experiments show progressively higher respiratory discrimination as the representation is enriched. Each block of Table 12 holds the other stream at its baseline, so no single row there is the full model, and the two baselines differ: the cardiorespiratory block runs the 188 neural descriptors without the pretrained embedding, while the neural block runs the 14 engineered cardiorespiratory descriptors. Consequently, the two blocks report different respiratory AUC for what both call an engineered-feature reference (0.698 against 0.685), and the adopted cardiorespiratory row reads 0.740 staging accuracy against the headline 0.739, which adds the pretrained embedding on top of it. The upper block replaces the 14 engineered cardiorespiratory features with progressively richer representations of the same underlying signal. Every step is significant, and the ordering identifies where the limitation lies. A *random* nonlinear expansion of the same 14 numbers recovers 0.018 AUC, a gain obtained without additional input information and from added capacity alone, indicating that the summary features leave nonlinear structure unrepresented. A random linear projection of the raw waveform adds a further 0.016, so part of what the features discard is recoverable without learned parameters. The learned convolutional encoder reaches 0.782, 0.084 above the engineered features.

These results indicate that respiratory performance is primarily constrained by the representation of the cardiorespiratory stream; the downstream detector is held fixed throughout the comparison.

The same representation search also evaluated MOMENT [52] as a frozen time-series encoder for the cardiorespiratory stream. Pretrained weights yield a respiratory AUC of 0.808, while randomly initialised frozen weights yield 0.795, with comparable staging accuracy in both cases (Table 12). The randomly initialised variant is also the time-series encoder selected most often by the nested procedure in Section 5.2. These controls support the representation-level finding across alternative high-capacity encoders. The learned convolutional encoder is retained in MM-Net as the compact implementation: it contains 0.155 million trainable parameters, is trained end to end with the rest of the network, and introduces no external model dependency, compared with the 341 million frozen parameters of MOMENT.

On the neural side, a frozen pretrained encoder complements the expert features without replacing them. The lower block of Table 12 varies the neural representation with the cardiorespiratory stream held at the engineered features. Used alone, the frozen encoder is significantly *worse* than the 188 features (-0.029, Holm $p = 0.008$): pretraining on healthy scalp EEG does not by itself substitute for physiology chosen for this task. Concatenated with the features

Table 12
Comparison of cardiorespiratory and neural representations (frozen-encoder configuration). Each block varies one representation while holding the other stream fixed; values are mean ± SD across thirty fold-seed runs, with Holm-corrected comparisons against the engineered-feature baseline. † denotes MOMENT-1-large.

Representation	Variant	Resp. AUC ↑	Staging acc ↑	Δ	Holm p
Cardiorespiratory	14 engineered features	0.698 ± 0.051	0.736 ± 0.014	—	—
	random nonlinear expansion of the 14	0.715 ± 0.047	0.737 ± 0.018	+0.018	0.012
	random linear projection of raw signal	0.731 ± 0.044	0.735 ± 0.021	+0.034	0.012
	learned CNN, 0.155 M (adopted)	0.782 ± 0.040	0.740 ± 0.019	+0.084	**0.010**
	random-init. time-series encoder†	0.795 ± 0.045	0.737 ± 0.017	+0.097	0.010
	pretrained time-series encoder†	$\mathbf{0.808 \pm 0.042}$	0.734 ± 0.018	+0.110	0.010
Neural	188 engineered features	0.685 ± 0.041	0.741 ± 0.016	—	—
	frozen pretrained encoder alone	0.681 ± 0.033	0.712 ± 0.033	−0.029	0.008
	features + pretrained (adopted)	0.690 ± 0.037	$\mathbf{0.749 \pm 0.018}$	+0.008	**0.006**
	features + two pretrained encoders	0.684 ± 0.043	0.745 ± 0.018	+0.004	0.23

it adds +0.008 accuracy (Holm $p = 0.006$), and a second pretrained encoder on top of the first adds nothing further (-0.004, $p = 0.56$).

These results support a narrow interpretation: expert physiology and self-supervised pretraining encode partly different information, so their combination outperforms either alone; the complementarity, however, saturates after one pretrained encoder, which does not support stacking several foundation models.

5.16. Integrating the Sleep-Adapted Encoder

Table 13 builds MM-Net from the frozen-encoder configuration one change at a time, every row on the same folds and seeds. Uniform class weights and cross-attention fusion contribute +0.009 together. The largest change comes from the encoder. A sleep-adapted encoder whose training embeddings are cross-fitted but not aligned moves the three-seed accuracy by only +0.002 ($p = 0.63$) and falls below the frozen encoder in five folds of ten, by up to 0.023 in the fold where the cross-fitted and full encoders agree least (embedding cosine similarity 0.53). Aligning the embedding spaces turns the same encoder into a gain of +0.014 (0.755 → 0.769, $p = 0.020$), and of +0.015 for the single model in nine folds of ten ($p = 0.027$). The complete MM-Net adds three elements that serve REM and the respiratory head more than overall staging: the ocular channels in the encoder, which raise the encoder's own accuracy from 0.709 to 0.732, nasal pressure, and the epoch alignment of Section 3. With all three, staging is 0.763; an EEG-only encoder in the same model gives 0.765, which is indistinguishable ($p = 0.38$), and MM-Net uses the EEG and EOG encoder in every cohort of this study.

On its own, the sleep-adapted encoder reaches 0.732 on single epochs without a temporal model, against 0.647 for the frozen encoder read by logistic regression. The remaining components of MM-Net add to this performance. With the engineered descriptors, the cardiorespiratory stream and the temporal decoder, the same encoder yields 0.756 as a single model, +0.024 in eight folds of ten ($p = 0.014$); on ISRUC-I the corresponding gain is +0.012 ($p = 0.014$; Section 5.19). Figure 7 shows what adaptation changes in the representation: held-out epochs that the frozen encoder spreads across one mixed cloud form stage-specific clusters after adaptation, with Wake, REM and N3 separated (silhouette by stage $-0.045 \rightarrow 0.047$).

Table 13
From the frozen-encoder configuration to MM-Net on iSLEEPS, one change per row (upper block), and the complete model (lower block); same ten folds and three seeds throughout. p: Wilcoxon on fold means against the row marked ref.

Configuration	Single	3-seed	p
Frozen encoder, class weights, concatenation	0.738	0.749	
+ uniform class weights	0.744	0.754	
+ cross-attention fusion	0.747	0.755	ref.
+ sleep-adapted encoder, cross-fitted, not aligned	0.752	0.757	0.63
+ ridge alignment (7 channels)	**0.762**	**0.769**	**0.020**
MM-Net: EEG+EOG encoder, 8 channels, epoch-aligned	0.756	0.763	
with an EEG-only encoder	0.757	0.765	

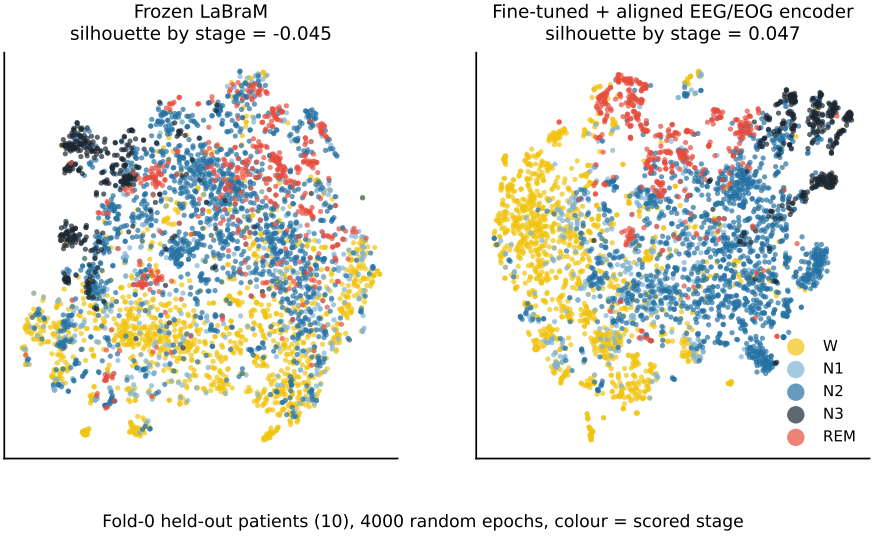


Figure 7: t-SNE [53] of 4000 epochs from the ten held-out patients of one fold, coloured by scored stage. Left: frozen LaBraM. Right: the sleep-adapted, aligned EEG/EOG encoder used in MM-Net.

5.17. The Two Loss Constants

Two constants in Section 4 were fixed by argument: the task balance $\lambda = 1$ and the positive weight $\pi = n_-/n_+$, which is 5.35 on this corpus. Equal coefficients do not imply equal influence, because the two terms differ in scale and curvature; $\lambda = 1$ is therefore a nominal choice. We therefore swept each in the frozen-encoder configuration with everything else at its reference setting, three seeds by ten folds per value, and evaluate both heads at every point, because the relevant outcome is the trade-off between the heads.

No trade-off is observed, and the effect of either constant is small. Across λ from 0.25 to 4 staging accuracy spans 0.003 and respiratory AUC 0.018, and across π from 1 to 11 they span 0.003 and 0.007; the fold-to-fold standard deviations are 0.019 and 0.038. Both heads attain their maximum at the same λ, which is inconsistent with a compromise setting: were $\lambda = 1$ balancing one task against the other, the two curves would peak on opposite sides of it.

Neither reference value is the maximum of those tested. Staging is highest at $\lambda = 2$ (0.7403) and at $\pi = 11$ (0.7422), and respiratory AUC at $\lambda = 2$ (0.7840) and $\pi = 8$ (0.7846). All of these margins are below a fifth of a fold standard deviation, so the sweep does not identify better settings; it establishes that the objective is insensitive to both constants across the range examined, and that fixing them a priori has no measurable cost on this corpus.

5.18. Calibration and Selective Prediction

Clinical use requires information on both the accuracy of the model and the reliability of its confidence. Two measurements address this for the frozen-encoder configuration, both computed on held-out epochs with the raw posteriors before HMM smoothing.

The model shows some overconfidence. Its expected calibration error is 0.076. We calculate this value using ten equal-width confidence bins weighted by occupancy. Bins containing fewer than fifty epochs are excluded from the calculation. The same calibration pattern appears across the confidence bins. For example, epochs with confidence scores above 0.9 are correct 88% of the time. The confidence scores are therefore most useful for ordering predictions by certainty. Section 4 explains why their numerical values differ from calibrated probabilities.

Abstention requires only a correct ranking of confidence. Applying split-conformal prediction with the fold's validation patients as the calibration set gives empirical coverage that tracks the target closely (0.894 at a 0.90 target, 0.943 at 0.95). At $\alpha = 0.10$ the model returns a single stage for 45.8% of epochs and is correct on 86.9% of those, against 59.1% on the epochs where it returns a larger set. Abstention therefore selects the more difficult epochs and concentrates on the stage that is hardest for human scorers: N1 receives a singleton set only 19% of the time, the lowest of any stage and the stage on which human scorers agree least.

The coverage estimates are subject to two limitations that do not directly affect the observed abstention behavior. The calibration patients are the same patients that stopped training, so their labels have already influenced which checkpoint was kept and they are not the independent calibration sample split-conformal theory assumes; the coverage we report is therefore empirical and carries no guarantee. In addition, epochs within a night are strongly dependent, so the exchangeable unit is closer to the patient than to the epoch; the reported values are therefore marginal rates across pooled epochs and do not represent coverage for a new patient or within any one stage. A calibration split disjoint from early stopping would resolve both issues; such a split was not used here.

5.19. External Validation on Independent Cohorts

Ten-fold cross-validation is patient-independent but remains within one corpus, one centre and one acquisition protocol. MM-Net is therefore evaluated in two further settings: trained and tested on independent cohorts under subject-independent cross-validation, with the encoder adapted, cross-fitted and aligned inside each fold exactly as on iSLEEPS; and trained on all 99 iSLEEPS patients and applied unchanged to external cohorts, with no re-fitted parameter, threshold or transition matrix (per-recording standardisation is applied, as in training).

Trained and tested on ISRUC-Sleep ISRUC-I is evaluated with ten subject-independent folds of 80 training, 10 validation and 10 test subjects, and ISRUC-II and III together with six folds that keep both nights of a subject on the same side. On ISRUC-I MM-Net reaches 0.803 accuracy, macro-F1 0.765 and κ 0.744 (Table 14). It is ahead of the frozen-encoder configuration in nine folds of ten (+0.018, $p = 0.004$), and the gain extends to respiratory detection (0.872 → 0.889 AUC) and to the AHI (ρ 0.523 → 0.590, mean absolute error 6.6 events/h). As on iSLEEPS, MM-Net is ahead of its sleep-adapted encoder used alone (0.792; +0.012 in eight folds of ten, $p = 0.014$). The reimplemented baselines, retrained on MM-Net's ISRUC-I folds with three seeds, reach 0.691 to 0.778; MM-Net is ahead of each of them in nine or ten folds of ten, by +0.025 over the corpus LSTM to +0.112 over Huttunen et al. (Holm $p = 0.018$, the minimum for nine comparisons), and ahead of Huttunen et al., the only other joint model, on respiratory detection as well (0.889 against 0.863 AUC). A model trained on healthy sleep and applied directly reaches 0.279, comparable to the 0.306–0.312 that healthy-trained checkpoints reach on iSLEEPS (Section 5.14). On ISRUC-II and III MM-Net reaches 0.725 over 26 nights and 0.779 on the ten healthy subjects of ISRUC-III.

Healthy sleep On Sleep-EDF, under ten subject-independent folds, MM-Net reaches 0.864 accuracy (macro-F1 0.791, κ 0.810), above every single-channel architecture reimplemented on the same recordings (0.791–0.835, Table 5). Sleep-EDF carries no cardiorespiratory channels, so this result exercises the neural stream alone. Applied without retraining, the frozen-encoder configuration trained on iSLEEPS reaches 0.794 accuracy (κ 0.718) on Sleep-EDF, above its in-domain score despite the unseen frontal montage. The same architecture therefore performs consistently across healthy sleep, a sleep-disorder cohort and stroke sleep.

Applied without retraining Table 15 applies MM-Net, trained only on iSLEEPS, to 151 external nights from 143 subjects. On the 100 ISRUC-I patients it stages at 0.730 and detects respiratory events at 0.833 AUC, and its AHI estimate reaches $\rho = 0.631$ (Figure 6). On UCD, an apnea cohort recorded on a different system with central EEG

Table 14
Trained and tested on ISRUC-Sleep. ISRUC-I: ten subject-independent folds (80/10/10 subjects), three seeds; baselines retrained on the same folds with three seeds (accuracy mean ± SD over folds). MM-Net is ahead of every baseline in nine or ten folds of ten (Wilcoxon, Holm $p = 0.018$). ISRUC-II+III: six folds keeping both nights of a subject together. Encoder alone: sleep-adapted EEG/EOG LaBraM, one epoch, no temporal model. Frozen-encoder configuration: class weights, concatenation, seven cardiorespiratory channels. AHI is reported for ISRUC-I, whose 100 patients span the severity range.

Model	Acc ↑	Macro-F1 ↑	κ ↑	Resp. AUC ↑	Resp. AP ↑	AHI ρ ↑	AHI MAE ↓
ISRUC-I (100 patients)							
StagingSeqNet (trained on Sleep-EDF)	0.279 ± 0.034	0.182	0.069	—	—	—	—
Huttunen et al. [13]	0.691 ± 0.044	0.655	0.603	0.863	0.417	—	—
TinySleepNet [8]	0.741 ± 0.057	0.710	0.666	—	—	—	—
U-Time [23]	0.743 ± 0.049	0.711	0.669	—	—	—	—
SleepTransformer [25]	0.751 ± 0.037	0.723	0.678	—	—	—	—
AT-BiLSTM [51]	0.755 ± 0.042	0.728	0.684	—	—	—	—
1D-ResNet-SE-LSTM [49]	0.758 ± 0.049	0.732	0.687	—	—	—	—
AttnSleep [7]	0.758 ± 0.047	0.717	0.685	—	—	—	—
LSTM (corpus)	0.778 ± 0.025	0.744	0.711	—	—	—	—
Encoder alone	0.792	—	—	—	—	—	—
Frozen-encoder configuration	0.786	0.752	0.722	0.872	0.447	0.523	6.8
MM-Net	$\mathbf{0.803 \pm 0.023}$	**0.765**	**0.744**	**0.889**	**0.477**	**0.590**	**6.6**
ISRUC-II+III (18 subjects, 26 nights)							
Frozen-encoder configuration	0.716	0.689	0.632	0.757	0.138	—	—
MM-Net	**0.725**	**0.692**	**0.641**	**0.794**	**0.197**	—	—
of which ISRUC-III (healthy)	0.779	0.747	0.714	0.867	0.260	—	—

Table 15
MM-Net trained on all 99 iSLEEPS patients and applied without retraining (three-seed, seven-channel variant without nasal pressure; AHI scale k from iSLEEPS). AHI is not reported for the healthy ISRUC-III subjects.

Cohort	Nights	Acc ↑	κ ↑	Resp. AUC ↑	AHI ρ ↑
ISRUC-I (sleep disorders)	100	0.730	0.641	0.833	0.631
ISRUC-II (sleep disorders)	16	0.681	0.568	0.783	0.518
ISRUC-III (healthy)	10	0.709	0.618	0.776	—
UCD (suspected sleep apnea)	25	0.607	0.463	0.841	0.653
iSLEEPS (in-cohort, reference)	99	0.763	0.656	0.825	0.765

only, respiratory detection reaches 0.841 AUC and the AHI estimate $\rho = 0.653$, with 60% of patients in the correct severity band (weighted κ 0.51); staging there (0.607) is limited by the absence of occipital EEG and by the unit-less recordings, neither of which affects the respiratory stream. Respiratory detection therefore transfers with little loss: 0.83–0.84 AUC on the two apnea cohorts against 0.825 in-cohort.

These results have one implication for deployment. Nasal pressure is cohort-specific: the eight-channel model, which reads it, transfers to ISRUC-I with 0.794 respiratory AUC and AHI ρ 0.449, whereas the seven-channel variant without it reaches 0.833 and 0.631, at the same staging accuracy. MM-Net therefore uses all eight channels when trained on a cohort and the seven-channel variant, reported in Table 15, when applied to a new acquisition system.

5.20. Model Complexity and Capability

Table 16 sets the proposed model beside representative alternatives on size and capability. MM-Net holds 2.85 M trainable parameters, with a 5.8 M-parameter sleep-adapted encoder trained in a separate stage; together they remain below the corpus's CNN-ResNet18 (∼11 M) and comparable to the larger single-channel stagers. MM-Net delivers the sleep stage, the respiratory-event score and the AHI from one pass over the recording, and none of the staging-only models, at any size, can supply the second and third outputs.

Table 16
Model size and output capability of the evaluated architectures, including whether each model supports sleep staging, respiratory-event prediction, or both.

Model	Input	Params ↓	Outputs
CNN-ResNet18 [22]	1 EEG	∼11 M	stage
DeepSleepNet [5]	1 EEG	∼21 M	stage
Raw multimodal CNN	14 ch	0.95 M	stage + apnea
MM-Net (proposed)	7 neural + 8 cardio	**2.85 M** + 5.8 M	**stage + events + AHI**

6. Discussion

This study shows that sleep staging, respiratory-event prediction and AHI estimation can be integrated within a single model for subacute ischemic-stroke PSG, and that the tasks retain a distinct physiological organization within that model. MM-Net reaches 0.763 sleep-staging accuracy ($\kappa = 0.656$), ahead of every architecture retrained on the same folds, together with a respiratory-event AUC of 0.825 and an AHI estimate that correlates with the annotated index at $\rho = 0.765$. The controlled interventions, performed in the frozen-encoder configuration, indicate how the two streams contribute: neural information carries most of the discriminative structure for sleep staging, whereas cardiorespiratory information carries the respiratory task, and joint optimization provides a smaller but consistent additional benefit across the two outputs. The resulting model therefore combines clinically related analyses through shared temporal context and retains task-specific physiological dependence.

The largest single gain comes from how the pretrained EEG encoder enters the model. Adapting LaBraM to sleep improves its standalone staging, yet the adapted encoder adds nothing measurable to the downstream model until its training embeddings are cross-fitted and aligned to the deployed encoder (Table 13). The failure mode, a downstream model trained on in-sample or misaligned representations, may apply to other pipelines that fine-tune a foundation model and then train a second model on its outputs, and the remedy costs K additional fine-tuning runs and a ridge regression per fold. The effect held on two independent cohorts: +0.014 on iSLEEPS with every other component fixed, and +0.018 for the complete model over the frozen-encoder configuration on ISRUC-I.

This organization extends previous joint sleep-analysis work to a stroke-specific setting. Huttunen *et al.* [13] demonstrated simultaneous sleep staging and respiratory-event scoring in a large suspected-obstructive-sleep-apnea cohort. Under the common iSLEEPS evaluation used here, their architecture reaches 0.599 staging accuracy against 0.756 for MM-Net, and on ISRUC-I, trained and evaluated on the same labels, 0.691 staging accuracy and 0.863 respiratory AUC against 0.803 and 0.889. The difference is important in the context of the present population, where the available montage, cohort size, neural physiology, and respiratory annotation differ from those of conventional sleep-apnea studies. The results indicate that joint PSG analysis in stroke benefits from representations and signal organization adapted to the characteristics of the target cohort.

The modality-ablation experiments provide the most direct evidence for this interpretation. In the frozen-encoder configuration, removing the cardiorespiratory stream reduces respiratory AUC to 0.654 while leaving staging essentially unchanged. Removing the neural stream produces the complementary response, reducing staging accuracy to 0.680 while respiratory discrimination is largely preserved. These interventions identify a strong task-specific dependence within the trained model. Cortical, ocular, and muscular information provides the principal basis for distinguishing sleep stages, while respiratory and cardiovascular signals provide the principal basis for identifying respiratory disturbance. The shared temporal decoder therefore connects two tasks whose immediate physiological evidence remains largely distinct.

The structure within the cardiorespiratory stream further clarifies how the respiratory output is formed. Respiratory effort is the strongest individual contributor, with its removal producing the largest channel-specific loss. Removing the complete cardiorespiratory stream, however, causes a much larger reduction than removing any single signal. This pattern indicates that respiratory information is distributed across partially overlapping measurements of breathing mechanics, oxygenation, and cardiovascular response. The airflow intervention gives this interpretation additional support: respiratory AUC remains 0.778 after airflow is removed, compared with 0.782 for the complete frozen-encoder model. The respiratory decision therefore remains supported by complementary physiological information even when the primary airflow signal is absent. This distributed dependence is also consistent with the stability of respiratory performance in recordings with incomplete cardiorespiratory montages.

The representation experiments identify where much of the modeling gain originates. Fourteen engineered cardiorespiratory descriptors reach an AUC of 0.698, while progressively richer representations of the same physiological signals recover additional performance, culminating in 0.782 for the learned raw-signal encoder. The nested model-selection experiment supports the same interpretation from a separate direction: several higher-capacity cardiorespiratory representations perform competitively across the outer folds. The important design principle is therefore the preservation of richer temporal information from the underlying respiratory and cardiac waveforms. CardioCNN implements this principle in MM-Net; across encoder configurations, the performance gain is associated with representation quality.

The neural stream follows a complementary pattern. Engineered descriptors form a strong standalone representation for sleep staging, and the frozen pretrained EEG representation adds information when combined with them. The engineered features reach 0.741 staging accuracy in the controlled representation experiment, and their combination with the pretrained representation reaches 0.749. This result suggests that sleep-specific physiological descriptors and large-scale EEG pretraining encode partly different aspects of the neural signal. The former contributes structure selected explicitly for sleep physiology, while the latter contributes representations learned from broader EEG data [9]. Their combination is well matched to the present setting, where the supervised stroke cohort is modest in size and fully end-to-end representation learning would place a greater burden on the available data. Adapting the pretrained encoder to sleep, with the ocular channels added, extends the same complementarity: the adapted encoder alone stages at 0.732, and within MM-Net, alongside the engineered descriptors, at 0.756 as a single model.

The matched single-task experiments show how these specialized streams benefit from joint temporal learning. Joint optimization raises staging accuracy from 0.733 to 0.739 and respiratory AUC from 0.774 to 0.782, with the same direction observed across most patient folds. The effect is consistent with the structure of the problem (these comparisons use the frozen-encoder configuration): sleep stage and respiratory disturbance arise from different immediate physiological measurements, yet both evolve within the same overnight sequence. Shared temporal modeling can therefore exploit common context while the modality-specific encoders preserve the information required by each task. The fusion experiments reinforce this interpretation. Concatenation and cross-attention combine the streams comparably, and the aligned cardiorespiratory pathway to the respiratory head provides an additional task-specific contribution. The results therefore indicate that model performance depends primarily on the quality and temporal alignment of the physiological representations, with limited additional benefit from increased fusion complexity.

The staging experiments also reveal the importance of evaluating sleep models within the population in which they will be used. Deep architectures retrained on iSLEEPS reach approximately 0.60–0.72 accuracy, while MM-Net reaches 0.763 with a macro-F1 of 0.664. The healthy-to-stroke transfer experiment shows an even larger shift: models achieving approximately 0.81–0.86 accuracy on Sleep-EDF reach only about 0.31 when applied directly to iSLEEPS without adaptation. Because population, montage, and acquisition conditions change together, this experiment is most appropriately interpreted as evidence of a substantial domain shift between conventional healthy-sleep data and subacute stroke PSG. The same MM-Net architecture reaches 0.864 accuracy under subject-independent evaluation on Sleep-EDF, 0.803 on ISRUC-I and 0.763 within the stroke cohort, further illustrating how strongly the observed performance depends on the target population.

The external evaluations provide complementary evidence that the learned representations remain useful beyond the development cohort. Trained on ISRUC-I, MM-Net reaches 0.803 accuracy and 0.889 respiratory AUC and is ahead of every retrained baseline on the same folds. Trained on iSLEEPS alone and applied without retraining, it stages the 100 ISRUC-I patients at 0.730, and on ISRUC-I and on UCD, an apnea cohort recorded on a different system, it detects respiratory events at 0.833 and 0.841 AUC and estimates the AHI with $\rho = 0.631$ and 0.653. The variation across these datasets reflects the expected influence of population, montage, class distribution, and signal availability, while the retained performance indicates that the learned physiological representations are not confined to a single recording environment. Nasal pressure was the input whose contribution depended most on the acquisition system in these experiments, and the seven-channel variant is the appropriate choice for transfer.

The respiratory output is trained at the 30-s temporal scale of staging, as a score for the onset of an event within each epoch. Its epoch-level score reaches 0.825 AUC and 0.487 average precision and is strongest on the most severe event types. Summed over decoded sleep and scaled once per fold, the same score estimates the AHI with $\rho = 0.765$ against the annotated index, closer than the clinical index itself agrees with those annotations ($\rho = 0.687$). The expected-count formulation is consistent with this agreement: it counts closely spaced events separately, where thresholding into runs would merge them, and it normalises by the decoded sleep time, as the clinical index does. A dedicated one-second

event detector does not improve the correlation, and averaging the two improves severity agreement. A usable AHI is therefore available from the staging model itself.

The patient-level analysis provides a final connection between the two tasks. Staging accuracy is 0.784 in patients with a normal AHI and remains at 0.754–0.764 in the mild, moderate and severe groups, without a significant trend, so the model stages the patients with the greatest respiratory burden nearly as well as those without it. Sleep architecture and respiratory disturbance remain distinct prediction problems at the signal level while sharing clinically relevant structure at the level of the overnight recording. This relationship supports their analysis within a common framework, particularly in a stroke population where both processes contribute to the physiological state of the patient during sleep.

The framework combines sleep-stage and respiratory-event information from the same PSG, preserves the physiological specialization of the neural and cardiorespiratory streams, and exploits their shared temporal context. Controlled interventions identify the signal groups supporting each output, representation experiments show how those signals are most effectively encoded, the encoder-integration experiments show how a pretrained EEG representation is brought into the model, and external evaluation demonstrates that the resulting representations retain utility across distinct PSG datasets. For subacute ischemic-stroke PSG, this provides a unified and physiologically interpretable analysis of sleep architecture, respiratory disturbance and its severity within the same overnight recording.

7. Conclusion

This study addresses the joint characterization of sleep architecture and respiratory disturbances in subacute ischemic stroke, a setting in which altered neural physiology and heterogeneous cardiorespiratory information challenge conventional sleep-analysis approaches. MM-Net demonstrates that sleep-adapted EEG representations can be integrated into a multimodal temporal framework through cross-fitting and embedding alignment, improving sleep-stage discrimination while supporting concurrent respiratory-event detection.

The findings establish two complementary aspects of the proposed approach. First, the integration of adapted neural representations provides a means of addressing the training–inference mismatch encountered when pretrained encoders are incorporated into downstream physiological models. Second, controlled interventions reveal distinct roles for neural and cardiorespiratory information within the joint framework, showing that shared temporal modeling can accommodate physiologically specialized prediction tasks.

Patient-independent evaluation and experiments across independent sleep cohorts support the applicability of this approach beyond a single recording population. Together, these results establish a computational framework for joint analysis of neural and respiratory physiology after ischemic stroke, while providing a foundation for further investigation of representation consistency and multimodal generalization in clinical sleep analysis.

Data and Code Availability

The iSLEEPS corpus is publicly available (Zenodo, DOI 10.5281/zenodo.14873844; and the india-data.org portal) as described by Maiti *et al.* [21]. ISRUC-Sleep [44], the UCD database [45] and Sleep-EDF [42] are public. Our code, the per-fold configurations, engineered features, the corrections applied to the external cohorts, and reproduction notebooks are released at `https://github.com/wshuv-o/MM-Net`.

Ethics

The iSLEEPS data collection was approved by the NIMHANS Institutional Ethics Committee (No. NIMHANS/34th IEC (BS&NS DIV.)/2022, dated 05.02.2022) [21]. This work is a secondary analysis of the de-identified public release and required no additional enrollment.

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